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Ground State Hyperfine Splitting At The Alpha Experiment At Cern

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Ringraziamenti

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Abstract

Since Dirac first theorized antimatter in 1928 and the discovery of the positron by Anderson in 1932, antimatter has been a subject of study in physics, particularly for investigating fundamental symmetries in nature, such as charge-parity-time reversal symmetry (CPT). After more than 100 years, antimatter is still an area of active interest, and studied both at large scale experiments, such the one operating at the Large Hadron Collider (LHC), and in smaller-scale experiment, such those conducted at the Antimatter-Factory at CERN, including the ALPHA experiment (*Anti- hydrogen Laser physics Apparatus*). The ALPHA experiment focuses on producing, confining and conducting spectroscopy as well as gravitational studies on antihydrogen atoms. Those atoms, produced by merging positron and antiproton plasma clouds in a Penning-Malmberg trap, are magnetically confined using superconducting solenoid and octupole magnets. Once a sufficient number of antihydrogen atoms have been accumulated in the trap, spectroscopic experiment are performed. These experiments induce quantum transitions, causing the atoms to change from trappable to untrappable states, which leads to their annihilation on the trap walls. These resulting annihilation events are detected by a silicon vertex detector, and constitutes the experimental spectra, from which the properties of the energy levels of antihydrogen atoms can be deduced. This thesis is dedicated to presenting the analysis and recent results achieved by the ALPHA collaboration, that has measured the hyperfine splitting of the ground state of antihydrogen with a remarkable precision of 4 parts per million (ppm), marking a significant advance in antimatter research. The ground state has been measured in a non-zero magnetic field configuration, where the degeneracy of the 1S energy level is fully resolved by the 1 T field. Of the four hyperfine spectral lines, which differ according to the relative spin orientations of the positron and antiproton, only two correspond to trappable states. The hyperfine levels are measured by inducing positron spin-flip transitions from trappable to untrappable states using microwave radiation directed into the trap, at increasing frequency steps. The analysis has consisted on developing the analytic model to interpolate the data, and estimate statistical and systematic uncertainties

Chapter 1

Introduction

1.1

Chapter 2

The Alpha Experiment

2.1 Antiproton decelerator facility at CERN (AD)

2.2 The ALPHA apparatus

2.3 The positron Accumulator

2.4 The ALPHA2 Penning-Malmberg Trap

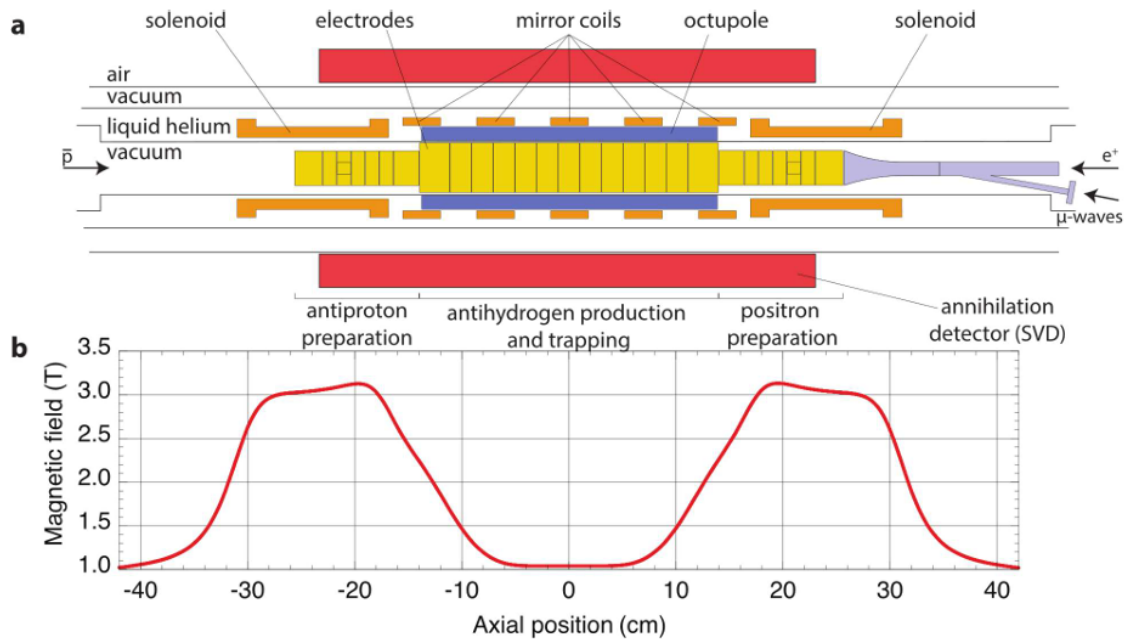


Figure 2.1: •

2.4.1 Electron Cyclotron Resonance Technique (ECR)

2.5 Synthesis of Antihydrogen

2.6 Detection of Antimatter

2.6.1 The ALPHA2 Silicon Vertex Detector

hyperfine splitting¹

¹1.

Chapter 3

Hyperfine Splitting of The Antihydrogen Ground State

This chapter recalls a bit of theory about the energy levels of antihydrogen and focuses on the lines that have been measured at ALPHA. Important to mention the zeeman broadening, since we are working at non-zero field, and the experimental protocol of the measurement. Things to mention are ECR, microwave hardware, magnetic field map. Then a description of the experimental protocol, magnetic field decay

The hyperfine structure is defined as a small shift (roughly 2000 times smaller than that relative to the fine structure) in the electronic energy level of atoms and molecules. This shift originates from the electromagnetic interaction of the electron magnetic momentum and the magnetic/electric multipole moment of the nucleus. Although the shift is smaller compared the magnitude of the energy level gap and other effects, such as the fine structure of the atoms, it has an important interest in atomic and astrophysics as well as in antimatter studies. From the atomic physics point of view, the transition between the 1s hyperfine levels of hydrogen is an important quantity than has been measured with a remarkable precision of 2 mHz ([6]), corresponding to a relative precision of 1.4×10^{-12} . In nuclear physics, transitions between hyperfine energy levels offers the possibility of investigating the internal structure of the proton and other nuclei and provide test of QED [7]. From the astrophysical point of view, the 1s hyperfine splitting corresponds to the 21 cm line, which is transparent to galactic medium and has been extensively used to map hydrogen density of galaxies, galaxy rotation curves, redshift effect [4] and to test variation of fundamental constants (fine structure constant and g factor of the proton) over time [8]. In the antimatter field, this spectral line can be used to test CPT symmetry, by repeating the same measurement on antihydrogen. Currently, two experiment at the antimatter factory at Cern have the capability to perform this measurement as part of their physics programs: the ASACUSA collaboration [5], that aims to measure the hyperfine splitting on a antihydrogen beam (on-flight measurement) and the ALPHA collaboration, which has achieved a measurement on 2017 [1] and [inserisci la citazione quando viene pubblicata](#). The present chapter will give an introduction to the physics underlying the measurement and outline the experimental protocol followed in 2023/2024, that has improved the uncertainty by a factor $\simeq 50$. The following derivation follows the approach in [3, cap. 12]. Only the case of the hyperfine level of the ground state, which are of particular relevance to this thesis, will be discussed; certain details will be omitted, to not burden the discussion

3.1 Hyperfine Structure

This section contains a brief derivation of the physics behind the hyperfine levels of antihydrogen. Present a brief derivation of the physics

The hydrogen atom is a bound state of a positron and an antiproton. The base Hamiltonian of this system, neglecting all higher-order terms, takes the form of $H_0 = \frac{p^2}{2\mu} + V(r)$ where μ is the reduced mass of the anti-proton/positron system, and $V(r)$ is the central static electromagnetic potential. When considering relativist effects, several high-order terms appears, which are encompassed in the equation¹:

$$H = m_e c^2 + \frac{p^2}{2m_e} + V(r) - \frac{p^4}{8m_e^3 c^2} + \frac{1}{2m_e^2 c^2} \frac{dV(r)}{r dr} L \cdot S + \frac{\hbar^2}{8m_e^2 c^2} \Delta V(r) + W_{hf} + \dots \quad (3.1)$$

the content of this full hamiltonian can be summarized as:

- $-\frac{p^4}{8m_e^3 c^2}$ correspond to the first relativistic energy correction, and can be easily obtained by expanding in series the relativistic energy $E = c\sqrt{p^2 + m_e^2 c^2}$
- $+\frac{1}{2m_e^2 c^2} \frac{dV(r)}{r dr} L \cdot S$ this is the term responsible of the fine structure of the atoms, and correspond to the coupling of the positron magnetic moment ($M_s = \frac{qS}{m_e}$) with its angular momentum L (spin-orbit interaction).
- $\frac{\hbar^2}{8m_e^2 c^2} \Delta V(r)$ is the Darwin term, that arises when we consider non-local interaction between the electron and the Coulomb field of the nucleus.
- W_{hf} is the term of the Hamiltonian which describes the hyperfine interaction

The first three additional terms causes the fine structure of the atoms, and have all α^2 order of magnitude. The term W_{hf} has the following expression:

$$W_{hf} = -\frac{\mu_0}{4\pi} \left(\frac{q}{m_e r^3} L \cdot M_I + \frac{1}{r^3} [3(M_s \cdot \hat{n})(M_I \cdot \hat{n}) - M_s \cdot M_I] + \frac{8\pi}{3} M_s \cdot M_I \delta(r) \right) \quad (3.2)$$

This expression contains new quantities that weren't introduced before: M_I is the magnetic moment of the nucleus, $\hat{n}$ is the unit vector connecting the antiproton to the positron, μ_0 is the magnetic permeability of vacuum. The first term is the equivalent of the spin-orbit interaction considering the nucleus spin, the second term is dipole interaction term and the last term is the Fermi contact term. This additional term in the Hamiltonian is the one causing the hyperfine structure of the levels, and it can be shown (look appendix) that the order of magnitude is 2000 times smaller that the fine structure terms **ricordati di fare i conti in appendice**. In general, the calculation of all these corrections to the energy levels and the interplay of the various physical effects is complex. However, for the specific case into consideration, numerous simplification of the various terms can be applied, making the calculations easier. For a 1s positron, the angular momentum $l = 0$, so for the term which explicit depend on L

¹In the full hamiltonian, $\mu = m_e$ has been used, assuming infinite mass for the proton

$$\langle \psi_{n=1,l=0,m_s,m_I} \rangle L \cdot S | \psi_{n=1,l=0,m_s,m_I} \rangle = 0 \quad (3.3)$$

The other terms, Darwin and the "relativist" term does not depend on either positron S and antiproton I spin. They only act as a coherent shift of the 4 sub-levels constituting the 1s energy level. Although not important for the following discussion, the shift is equal to (The derivation of this formula is left to the appendix):

$$-\frac{1}{8}m_e c^2 \alpha^4 \quad (3.4)$$

For W_{hf} , the first term depending on L is equal to zero for the same argument used above. The second term, the dipole-dipole interaction, is zero because of the spherical symmetry of the s-wave $\psi_{n=1,l=0,m_s,m_I}$. The only relevant term is the Fermi contact term. This term can be written as:

$$\langle \psi_{n=1,l=0,m_s,m_I} \rangle \left(\frac{2}{3} \mu_0 M_s M_I \delta(r) \right) | \psi_{n=1,l=0,m_s,m_I} \rangle = A \langle m_s, m_I \rangle I \cdot S | m_s, m_I \rangle \quad (3.5)$$

where A is equal to $\frac{e^2 g_p}{3\epsilon_0 c^2 m_e M_p} \frac{1}{4\pi} |\psi_{n=1,l=0}(r)|$. In the expression, the wave function evaluated at $r = 0$ has appeared in the calculation. This term link the value of the hyperfine splitting to nuclear physics, as the internal structure of the nucleus affects the value of $|\psi_{n=1,l=0}(r)|$. **The magnitude of this effect is about 33 part per million, as reported in [7].** It's not the goal of this thesis to present the state of the art about the various correction term. The value of A will be computed using the approximation of point-like antiproton. In this way, using $\psi_{n=1,l=0} = \frac{2}{a_0^3} e^{-\frac{r}{a_0}}$, the expression for A is:

$$A = \frac{4}{3} \frac{g_p}{\hbar^2} \frac{m_e}{M_p} m_e c^2 \alpha^4 \left(1 - \frac{m_e}{M_p}\right)^{-3} \quad (3.6)$$

Including these simplifications, W_{hf} can be written as $AS \cdot I$, that correspond to the additional term of the zero order hamiltonian. The eigenvalues of this system are easy to compute considering the total momentum $F = (S + I)$. Since $[S \cdot I, F] = 0$, hamiltonian can equivalently take the the form:

$$H = H_0 + \frac{A}{2}(F^2 - I^2 - S^2) \quad (3.7)$$

In this basis of $\{|F, m_F\rangle\}$, H is already diagonal. The energy levels of the system can be directly computed, and are the following:

$$H\psi_{n=1,l=0} |F, m_f\rangle = \left(E_0 + \frac{A\hbar^2}{2}(F(F+1) - I(I+1) - S(S+1)) \right) \psi_{n=1,l=0} |F, m_f\rangle \quad (3.8)$$

$$F = 1 \Rightarrow E_0 + \frac{A\hbar^2}{4} \quad (3.9)$$

$$F = 0 \Rightarrow E_0 - \frac{3A\hbar^2}{4} \quad (3.10)$$

It is meaningful to express the energy states in terms of the positron/antiproton spin orientation. Using the Clebsch-Gordon coefficients, each state of $F = 1$ and $F = 0$ correspond to:

$$|F = 1, m_f = 1\rangle = |+, +\rangle \quad (3.11)$$

$$|F = 1, m_f = 0\rangle = \frac{1}{\sqrt{2}}(|+, -\rangle + |-, +\rangle) \quad (3.12)$$

$$|F = 1, m_f = -1\rangle = |-, -\rangle \quad (3.13)$$

$$|F = 0, m_f = 0\rangle = \frac{1}{\sqrt{2}}(|+, -\rangle + |-, +\rangle) \quad (3.14)$$

The effect of W_{hf} term is to split the energy levels of the triplet $F = 1$ and the singlet $F = 0$. This split correspond to hyperfine structure of the $1s$ energy level at zero magnetic field, and it is then equal to $A\hbar^2$. The frequency of the transition, substituting the value of the constant A in the formula is equal to $f = \frac{\nu}{2\pi} = \frac{A\hbar}{2\pi} = 1426849400.435 Hz$. The most precise measurement is $1420405751.768(2) Hz$, which is 0.45% off the prediction presented here.²

3.2 Zeeman Effect, Magnetic Field Decay

Now, from the calculations presented before, introduce the Zeeman Broadening and the degeneracy resolution into 4 different sublevels. Introduce the magnetic field decay, that will be an aspect to be treated in Experimental protocol

The results achieved in the previous section does not fully represents the experimental conditions at the ALPHA experiment. The main difference is that ALPHA operated with a 1 T field, in order to contain antihydrogen atoms one they have been synthesized in the Penning-Malmberg trap (see [richiamo alla sezione introduttiva](#)). The external magnetic field introduce an additional term in the hamiltonian, which resolve the degeneracy of $F = 1$. The hamiltonian in presence of a magnetic field is:

$$H = H_0 + \frac{A}{2}(F^2 - I^2 - S^2) - B_0 \cdot (M_s + M_I) \quad (3.15)$$

That can be expressed (using $M_s = \frac{q}{m_e}S$ and $M_I = -\frac{qg_p}{2M_p}$) in terms of $\omega_0 = \frac{-q}{2m_e}B_0$ and $\omega_n = \frac{q}{2M_p}g_pB_0$ as:

$$H = H_0 + \frac{A}{2}(F^2 - I^2 - S^2) + 2\omega_0 S_z + \omega_n I_z \quad (3.16)$$

the term ω_0 corresponds to the Larmor frequency precession, and the terms ω_n is the equivalent for the antiproton spin. In this derivation, we can approximate $\omega_n = 0$, since its value is 2000 smaller than ω_0 . The size of this new term is directly proportional to the applied magnetic field, as a consequence also the energy levels will depend on the value of the magnetic field. Another consequence of 3.16 is the fact that the basis $\{|F, m_f\rangle\}$ is no more a diagonal basis, since F does not commute with S_z . To solve the system and obtain the eigenvalues, the operator to resolve takes the form $AI \cdot S + 2\omega_0 S_z$. Firstly, we can evaluate the effect of S_z on base vectors of $\{|F, m_f\rangle\}$ written in the form of 3.11. The result is:

²This simple theoretical prediction is the result of many approximation. Complete calculations should involve qed correction, internal structure of the antiproton and other effects. A more comprehensive review is in [7]

$$S_z |F = 1, m_f = 1\rangle = \frac{\hbar}{2} |F = 1, m_f = 1\rangle \quad (3.17)$$

$$S_z |F = 1, m_f = 0\rangle = \frac{\hbar}{2} |F = 0, m_f = 0\rangle \quad (3.18)$$

$$S_z |F = 1, m_f = -1\rangle = -\frac{\hbar}{2} |F = 1, m_f = -1\rangle \quad (3.19)$$

$$S_z |F = 0, m_f = 0\rangle = \frac{\hbar}{2} |F = 1, m_f = 0\rangle \quad (3.20)$$

$$(3.21)$$

The application of S_z acts to "mix" the subspace spanned by the two vectors $|F = 1, m_f = 0\rangle$ and $|F = 1, m_f = 0\rangle$. With these expressions, all the elements to write the new terms of the hamiltonian in matrix notation are available, allowing for the calculation of the new eigenvalues by solving $\det(\mathbf{V} - \lambda \mathbf{I})$. the basis will be represented with the following notation : $|F = 1, m_f = 1\rangle =^T [1, 0, 0, 0]$, $|F = 1, m_f = -1\rangle =^T [0, 1, 0, 0]$, $|F = 1, m_f = 0\rangle =^T [0, 0, 1, 0]$, $|F = 0, m_f = 0\rangle =^T [0, 0, 0, 1]$. In this basis, the operator takes the form:

$$\begin{bmatrix} \frac{A\hbar^2}{4} + \hbar\omega_0 & 0 & 0 & 0 \\ 0 & \frac{A\hbar^2}{4} - \hbar\omega_0 & 0 & 0 \\ 0 & 0 & \frac{A\hbar^2}{4} & \hbar\omega_0 \\ 0 & 0 & \hbar\omega_0 & -\frac{3A\hbar^2}{4} \end{bmatrix} \quad (3.22)$$

In this notation, it's immediate to notice that the states $|F = 1, m_f = 1\rangle$ and $|F = 1, m_f = -1\rangle$ are still eigenvector of the matrix. So the energy levels are:

$$|F = 1, m_f = 1\rangle = |+, +\rangle \rightarrow \frac{A\hbar^2}{4} + \hbar\omega_0 = \frac{A\hbar^2}{4} - \hbar|\omega_0| \quad (3.23)$$

$$|F = 1, m_f = -1\rangle = |+, +\rangle \rightarrow \frac{A\hbar^2}{4} - \hbar\omega_0 = \frac{A\hbar^2}{4} + \hbar|\omega_0| \quad (3.24)$$

For the other two eigenvalues, the equation $\det(\mathbf{V} - \lambda \mathbf{I})$ must be solved, for the 2×2 sub-matrix

$$\det(\mathbf{V} - \lambda \mathbf{I}) = \left(\frac{A\hbar^2}{4} - E\right)\left(-\frac{3A\hbar^2}{4} - E\right) - \hbar^2\omega_0^2 = 0 \quad (3.25)$$

The solutions are:

$$-\frac{A\hbar^2}{4} \pm \sqrt{\left(\frac{A\hbar^2}{2}\right)^2 + \hbar^2\omega_0^2} \quad (3.26)$$

Hence, the complete set of eigenvalues has been determined. In scientific literature is common to present a the plot of the energy levels as a function of the magnetic field, known the Zeeman diagram of the system. The Zeeman diagram is shown in figure 3.1

As explained in the introduction [richiamo all'introduzione](#), the only trappable states are the c and d, which are the "low-field seeking" states. In high field regime, that is the case of the ALPHA experiment (magnetic field at $\simeq 1$ T), the hyperfine splitting is defined

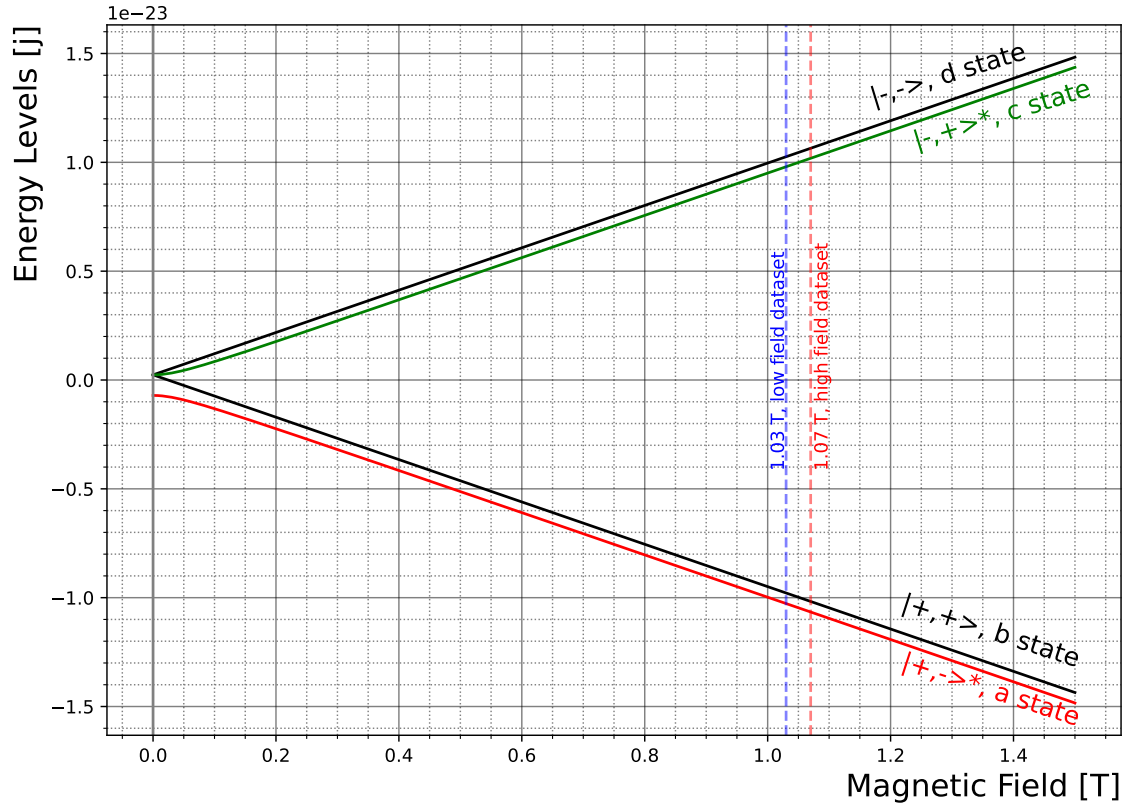


Figure 3.1: Zeeman plot of the 4 hyperfine Levels of the 1s state. The blue and red dotted lines denote the magnetic field configuration corresponding to the two dataset used for the measurement. Each of the four lines is labeled accordingly to its spin configuration at high magnetic field regime. The d and b states correspond exactly to the parallel and antiparallel spin alignments (up-up and down-down) for the antiproton and positron. The c state and a state are mixed due to the presence of the external magnetic field, and only in the high field limit the states correspond to the up-down or down-up spin configuration.

as the frequency difference between the transition from the **d** state to the **a** state and the transition from the **c** state and the **b** state.

$$f_{da} - f_{cb} = [(E_d - E_a) - (E_c - E_b)] \frac{1}{2\pi\hbar} = \frac{A\hbar}{2\pi} \quad (3.27)$$

it is important to highlight that this relation is valid only if the two transitions are measured at the same magnetic field value. This point is particularly significant, as it introduces one of the key characteristics of the magnetic field in the experiment. In fact, the large superconducting solenoid in alpha, the *Carlsberg* magnet, produces a magnetic field that is affected by drift over time. The characterization of this drift is made through the ECR technique ([reference](#)), which measure the magnetic field along z axis of the trap. The ECR measurement are reported in figure 3.2, that shows the magnetic field value measured at 261.8 mm [explain why at this location](#) at different time during the experiment. While the experimental protocol will be discussed later, it is evident from the figure that there is an initial transient phase, during which the magnetic field remains constant or exhibits an upward drift (attributed to fine scale flux relaxation following the solenoid reset). This is followed by slower, long-term, downward linear drift of the field, due to decay of the of persistent currents in the external solenoid. The linear drift of the magnetic field is expected to be negative, causing a shift in the resonance of the transitions of about 70 kHz h^{-1} .

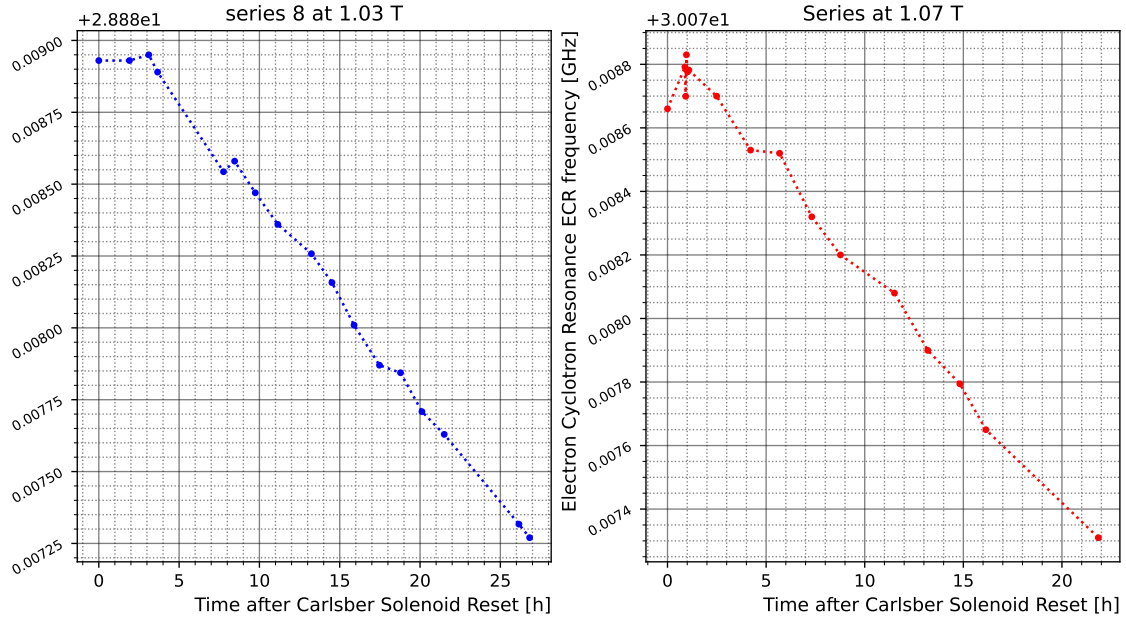


Figure 3.2: ECR measurement of the Magnetic field drift over time. The two series on points show the trend of the magnetic field over time from the most recent reset of the *Carlsberg* superconducting solenoid, for the two dataset used for the measurement, which differ for a small offset of the magnetic field at the trap center ($\simeq 0.04 \text{ T}$)

3.3 Experimental Protocol

Description of the experimental protocol: stacking, measurement scan from lower to higher frequencies ecc. Things to mention are:

- *destructive measurement: depletion effects*
- *it's a measurement by difference! It's important to mention it!*
- *Magnetic field profile*
- *step duration, clearing...*
- *qualitative interpretation of the shape: depletion effect, resonant volume ecc...*

This section provides a detailed description of the experimental protocol followed during data collection. The experimental data for spectroscopy measurement at ALPHA are constituted by looking at number of annihilating atoms over time. Depending on the different lines that the experiment aims to measure, various lasers, as the 243 nm for 1s-2s line, the 121 nm for the Lyman- α or microwave light are used to kick-out atoms, which annihilate on the trap walls. The physics process involved can be different: for the 1s-2s spectroscopy, ionization and momentum transfer from the laser light to the atoms plays the main role, while for the hyperfine measurement, the positron spin flip from a trappable to un-trappable state is the main physics mechanism for the ejection of atoms.

3.3.1 Microwave Induced Annihilation

To probe the **cb** and **da** transitions, microwave light is delivered into the trap environment via the wave guide depicted in figure 2.1. The frequency of the light is tuned between the range of 28 GHz-31 GHz, selected based on the corresponding operating point from the 1 T magnetic field. The number of atoms available for spectroscopy at the ALPHA experiment have increased from few tens to several hundreds/thousands atoms [2]. At first, these atoms are evenly split between the **c** and **d** states, while **b** and **a** states remain unpopulated, since atoms produced in these states rapidly (≈ 10 ms) escapes and annihilates on the trap wall upon creation. The transitions are probed by injected microwave light inside the trap cavity at varying frequencies, in increasing steps of 5 kHz and 8 s time duration. By varying the frequency, the microwave light can drive a positron spin-flip of an atom, modifying its quantum state. Once an atom ends up in an un-trappable state it annihilate on the trap wall, and the annihilation products (typically pions) are then detected by the Silicon Vertex Detector (SVD) surrounding the trap electrodes. By properly tuning the frequency of the microwave light, it's possible to select the **cb** or the **da** transition, and record different spectra from which physical quantities can be inferred. The protocol used during the 2024 data taking comprises a number of 48 frequency steps, (for total of 240 kHz), followed by an high power phase, named "clearing", consisting of 16 frequency steps of the same duration, during which residual **c** state or **d** states atoms that did not undergo transition are removed. Strictly speaking, the duration of spectroscopy measurement was 512 s, during which several hundreds/thousands atoms are recorded.

3.3.2 Protocol of Spectroscopy Experiment

To measure the $f_{c \rightarrow b}$ and $f_{d \rightarrow a}$ transition frequency, approximately 1400-1600 atoms are synthesize and trapped from repeating positron/antiproton mixing cycles over a period of

≈ 1 h. During each mixing cycle, a total of 90/100 atoms are trapped in the 0.5 K magnetic potential well. During the microwave experiment a single **cb** followed by a **da** spectra are measured. To excite the **da** transition, the microwave frequency scan start at 1 420 390 kHz above the starting frequency of the previous **cb** transition. A pause lasting 300s divides the two spectroscopy phases, during which the microwave system is switched off to cool down the electrodes of the trap, with the goal of avoiding to deteriorate the vacuum conditions inside the trap cavity. A $f_{c \rightarrow b}$ and $f_{d \rightarrow a}$ measurement constitutes a single replica, which has been repeated consecutively 8 times. At the end of the 8 repetitions, an high statistic repetition, with several thousands of atoms, has been measured, and used as a 'control' sample for systematic studies. This procedure has been repeated twice, yielding two datasets that differ in the magnetic field setting used for the spectroscopy. One dataset was taken at 1.03 T and the other at 1.07 T [explain why this choice](#). These settings correspond to different on-axis field minima, achieved by adjusting the target current in the Carlsberg solenoid. An additional difference is that a high statistic measurement, performed before the 8 standard repetitions, was included only for the series at low magnetic field. The experimental procedure can be summarize as follows:

- Reset of the Carlsberg Magnet to the desired current
- Stacking phase
- ECR
- low power **cb** sweep to identify the frequency start
- ECR
- **cb** scan (48 steps) and clearing (16 steps)
- 300 s of wait to cool down electrodes
- **da** scan (48 steps) and clearing (16 steps)
- ECR

We will refer to these as low field and high field datasets, respectively. During the measurement, Electron Cyclotron Resonance technique has been employed in between physical runs to monitor the magnetic field.

It is important to clarify that, in this context, the experimental spectra refer to the annihilation counts recorded by the SVD over time. These spectra are influenced by the depletion of the total atomic population available during the microwave sweep, and are shaped by several others factors, as the temperature of the atom sample, the unknown standing wave pattern, Doppler broadening, magnetic field inhomogeneities and the natural linewidth of the transition. The interplay of all these effects collectively determines the observed spectral shape. However, since the relevant physical quantity is obtained from the difference between $f_{c \rightarrow b}$ and $f_{d \rightarrow a}$ transitions, the key point is the relative impact of these effects on the two transitions. In this sense, the "by difference" nature of the measurement is advantageous: systematic effects that affects both transitions similarly can partially, or even fully, cancel out.

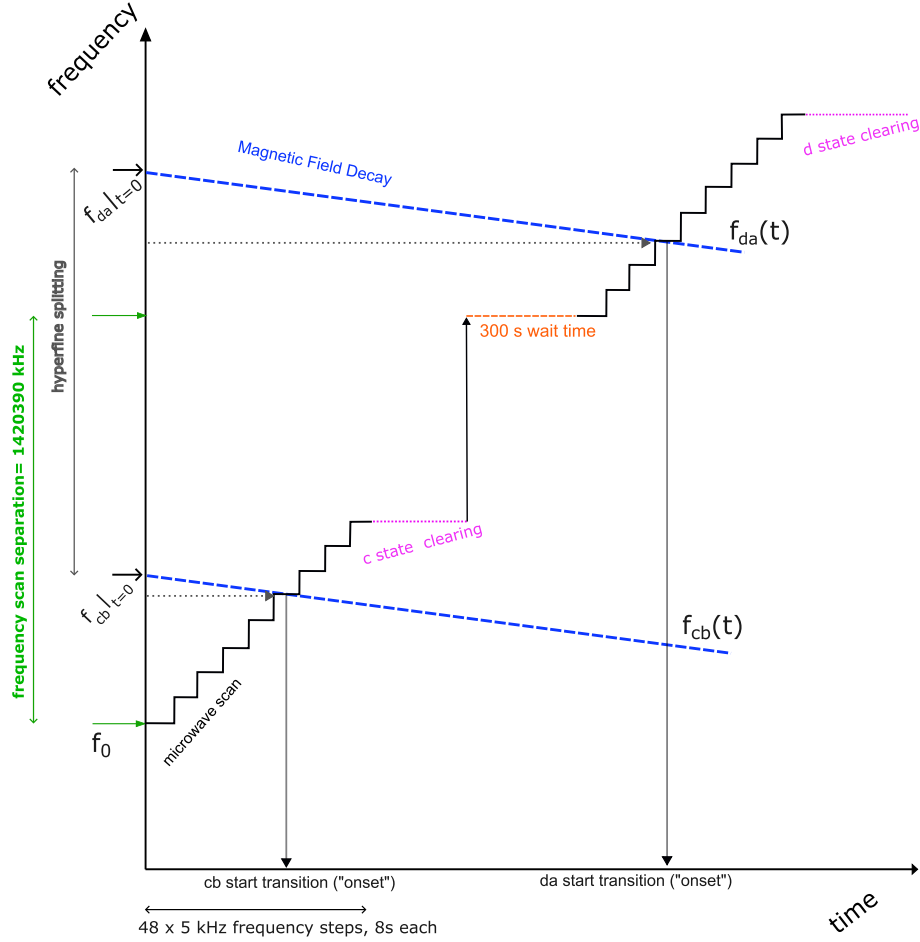


Figure 3.3: Scheme of a single experimental repetition. Each dataset consists of 8 such repetitions, with atoms stacked between each run for the next cycle. The initial frequency f_0 , the first microwave sweep of the $f_{c \rightarrow b}$ scan, is determined by a low power scan to locate the "onset" of the transition. For subsequent repetitions, the starting frequency of each sweep is located is adjusted by subtracting the expected magnetic field drift.

3.3.3 Magnetic Field Profile

Qui sei stato un po' sintetico. Sicuramente c'è da recuperare materiale aumentare questa parte

Special attention was devoted to the design of the magnetic field. The magnetic field was shaped to reduce the second order-derivative. Physically, in presence of a "steep" magnetic field, the atoms transverse the field minimum rapidly, thereby reducing their interaction time with the microwave field necessary to induce transitions. Instead for a flattened field profile, the region where the "resonant" frequency closely matches the injected microwave frequency is broader, which enhances the probability of transition. Figure 3.4 shows the magnetic field profile in the vicinity of the global magnetic field minimum. The magnetic field is deliberately depressed below the central mirror coil (by a 50 μT) to create a well defined absolute minimum. Neglecting all other effects that contributes to broadening, this result in a narrow and well defined resonance frequency. During the frequency scan, the microwave frequency probes transitions associated to higher value of the magnetic field, and the spectra are therefore subject to Zeeman broadening.

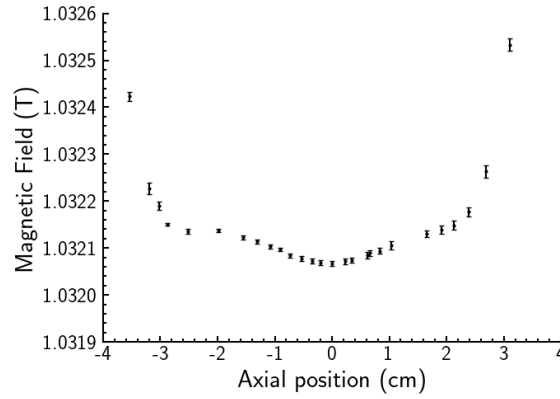


Figure 3.4: On-Axis magnetic field profile measured through ECR repeated at different Z position of the plasma. The resolution of the technique is 1 mm

The Zeeman defines also the order of the measurement: in principle one could measure the $f_{d \rightarrow a}$ transition first and the $f_{c \rightarrow b}$ as the second. But by doing so, some of the c-state atoms could also contribute to the $f_{d \rightarrow a}$ signal, since they can explore magnetic potential region that matches the resonance condition. While for the other order, the d-state atoms would need to explore a magnetic potential region below the actual field minimum.

3.3.4 Microwave System: Power and Duration

Talk about the setup of the microwave system, issues with the energy distribution and not very well defined resonance, introduce the microwave flat field for

The microwave power is produced by an analog signal generator (Agilent 8257D PSG) and subsequently amplified by a Miteq AMF-4B amplifier. The microwaves are delivered into the trap volume through a vacuum window and waveguides. Prior to the experiment, the injected power is adjusted to produce similar transition lineshapes with high statistic run, and matching the profiles of the c-to-b and d-to-a transitions. Variation in power delivered to the trap, as well as differences in the standing wave pattern can account for discrepancies in the shape between the two transitions, and therefore introduce a systematic error in the hyperfine measurement. The actual power in the trap and the

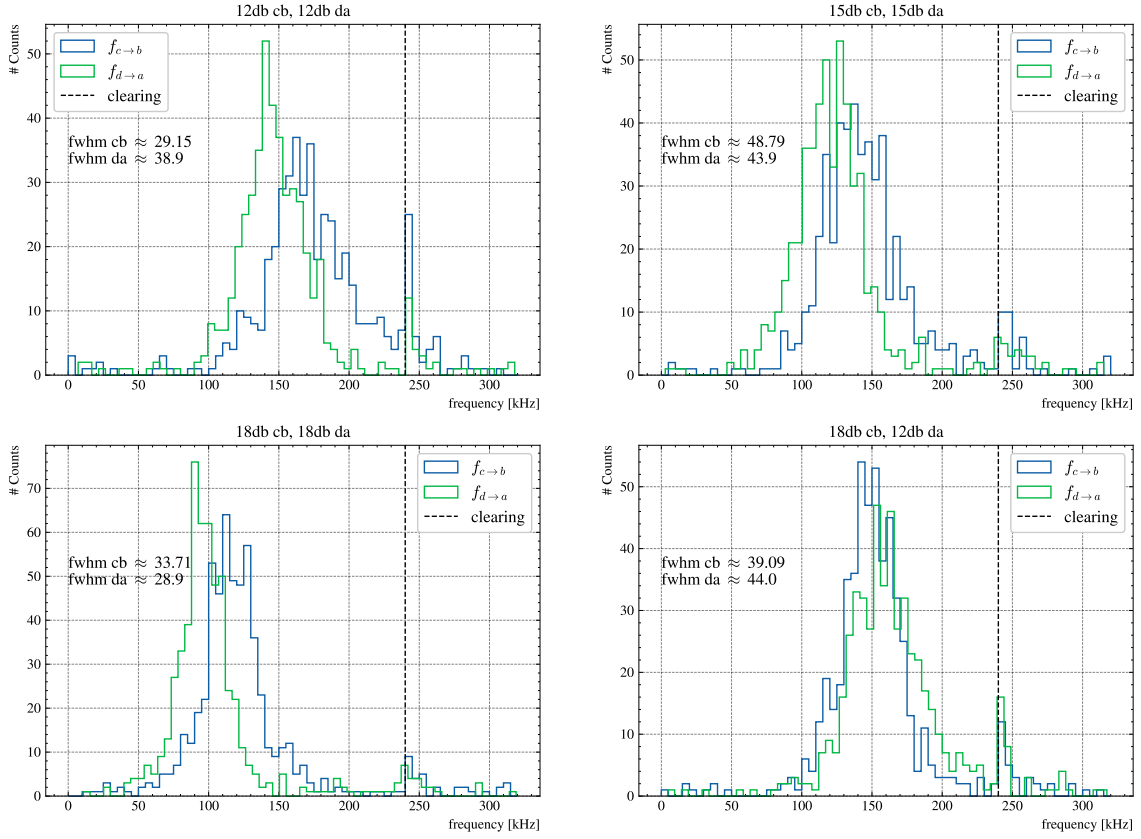


Figure 3.5: Lineshape collected with different nominal power of the microwave amplifier, for the set of data taken at 1.07 T, the frequency step is 5 kHz, with a pulse length of 8 s, *spiegare cosa sono 12db*

standing wave pattern are not precisely known, and are expected to vary depending on the frequency of the injected microwave light. Therefore, part of the data analysis is devoted to estimating the systematic uncertainty associated to these variations. This will be discussed in detail in the next chapter. To choose the microwave power for the experiment, a series of lineshape was measured while varying the microwave amplifier output power. The goal of this campaign was to balance the observed lineshapes and identify the power setting that yields the closest match between the $f_{c \rightarrow b}$ and $f_{d \rightarrow a}$ transition profiles. Some of the spectra collected at the high field are reported in 3.5

Regarding the step duration, that for the nominal datasets is 8 s, this has been selected by performing measurement with high statistic runs varying the time duration of microwave irradiation step, in the range of [1 s, 32 s].

The distributions in figure 3.6 have been interpreted as follows: when the microwave frequency approaches the value corresponding to the magnetic field minimum, atoms begin to be ejected from the trap and subsequently annihilated. The probability of ejection is proportional to the duration for which the atoms remains in the vicinity of their resonance condition while exposed to the microwave field. For shorter microwave pulse duration, some atoms capable of undergoing transitions at a given frequency may not be ejected due to insufficient pulse length. This accounts for a broader full width at half maximum fwhm observed for the 1 s case. Increasing the pulse duration leads to a decrease in fwhm, and

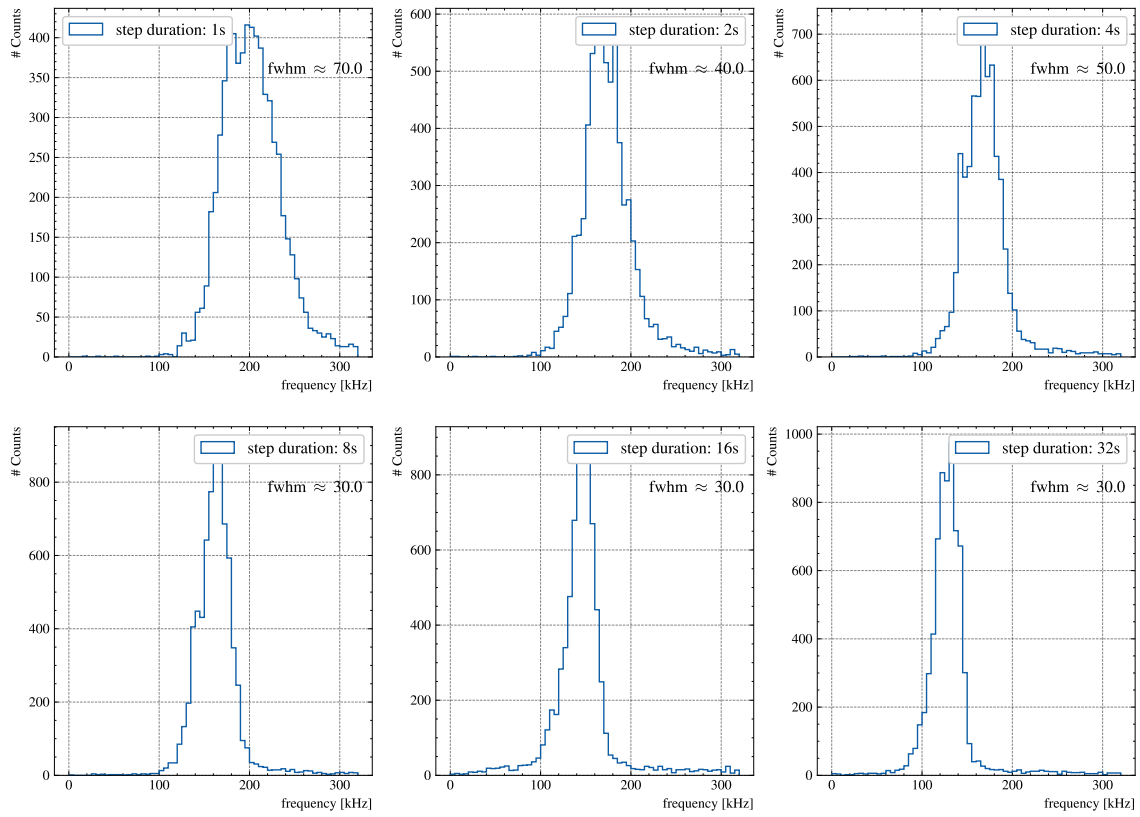


Figure 3.6: Collection of high statistic $f_{c \rightarrow b}$ lineshapes measured for various pulse length duration. The data shown are referring to the *Passed Cuts* selection. The starting frequency is expressed arbitrary to 0 kHz

from 8 s to 32 s the spectra converge to an *fwhm* of 30 kHz. The behavior indicates that for duration longer than 8 s, the pulse is sufficiently long for all atoms at each microwave step that are able to undergo a transition to do so, resulting in a "saturation" of the spectra lineshape. Following this interpretation, the step duration has been fixed to 8 s. Longer pulses does not offer any benefit in terms of width of the lineshape, on the other hand they contribute more to heating the trap electrodes, resulting in a worse vacuum condition and potentially more background due to annihilation of residual gas inside the trap.

3.4 Qualitative Interpretation of the Observed Spectra

In the previous section some $f_{c \rightarrow b}$ and $f_{d \rightarrow a}$ spectra have been presented. As anticipated before, the measured lineshape are the convolution of the several physical phenomena that broad the natural linewidth of each transition. A qualitative interpretation of the experimental lineshape can be offered by looking at the magnetic field profile in 3.4. As the microwave frequency get closer to the corresponding value at the minimum magnetic field, the a small percentage of the trapped atoms begin to annihilate (transition onset). At the beginning, only the atoms which pass through the magnetic field minimum are sensitive and can effectively undergo transition. While the microwave scan procedes toward higher frequency, the "resonant volume" increases, and so more atoms whose trajectory is not crossing the magnetic field minimum, are kicked out. Since the total number of atoms is fixed at the beginning of the experiment, the right part of the spectrum is dominated by depletion effects. The lineshape can be approximately divided in three section: a rising left tail, corresponding to probing in the proximity of the magnetic field minimum, a central part which describe the increase of the resonant volume as the scan proceed and a decreasing right tail, due to depletion effect.

Chapter 4

Data Analysis And Systematic Uncertainties

This chapter focuses on the statistical and mathematical framework used to analyze the data collected during the 2024 experimental run, related to the hyperfine splitting of the antihydrogen atom. The improvements in trapping efficiency, magnetic field design, colder energy distribution and characterization of the magnetic field drift have all contributed to an improvement of 10^2 compared to the first observation made by the alpha collaboration in 2017 [1]. In 2024, two complete set of data, with auxiliary measurement as ECR and high statistic control sample have been taken and analyzed. The analysis method is based on a combined fit of the transitions to extract the frequency marker/onset of each transition, and it has been developed with *ROOT* and *RooFit* packages, with a Maximum Likelihood minimization, using the frequentist approach. In a synthetic way, the analysis consists of a fit to the **cb** and **da** frequency distributions with an empirical lineshape model, common to all repetitions within a given series, but with some parameters which are allowed to differ for the two transitions, followed by a correction for the magnetic field decay since the last reset of the Carlsberg solenoid.

questa parte da inserire altrove, non qui Although tempting, these frequency markers are only related to the resonance frequency at a given condition, because of several broadening effects that will be discussed in this chapter. Because of its nature of "differential" measurement, the analysis focuses on identify those frequency markers without claiming anything about the resonance frequency of a given spectra alone. The output of the analysis consist on a common linear fit to the 8×8 frequency markers, correcting for the magnetic field drift and correlation effect to extract the physical quantity.

The analysis is developed using *ROOT* and *RooFit* data analysis package. The analysis software was developed in python with the *PyROOT* interface. The minimization routine that has been used during the Maximum Likelihood Fit (*MLE*) is *MINUIT2*

4.1 Background Model

In the analysis, a flat distribution for the background is considered. The main source of background is constituted by cosmic event, mainly muons, that are wrongly recorded as annihilation from the Silicon Vertex Detector. The cosmic background has been decrease by employing multi-variable-analysis techniques, that employs information of the event structure to reject background.

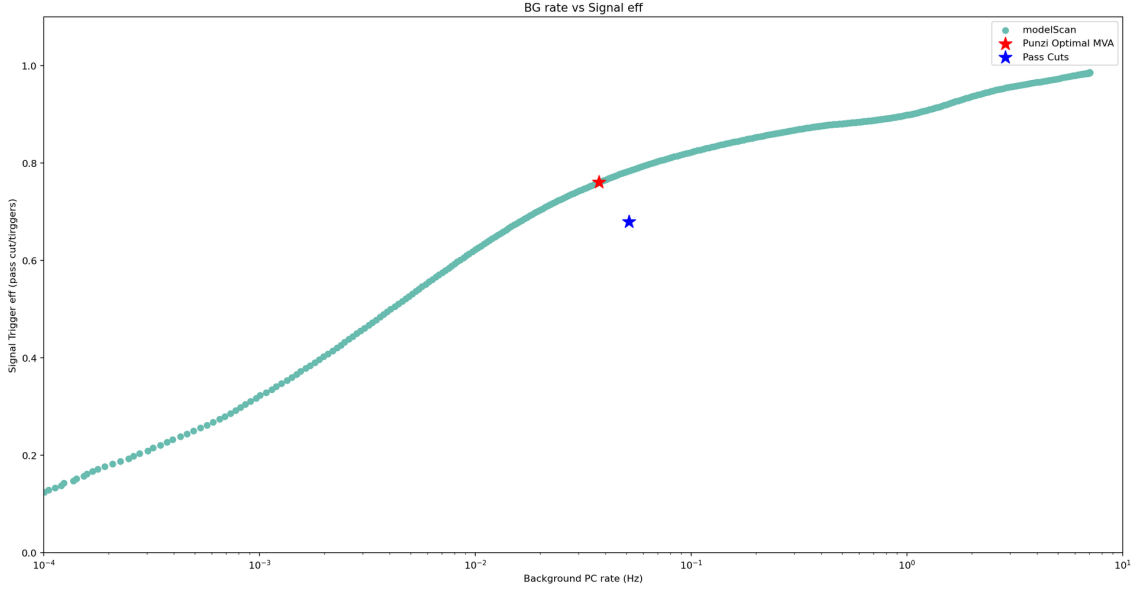


Figure 4.1: MVA scan plot. The cyan curve represents scan of the MVA model varying the selection criteria. The red star is the optimal point corresponding to the minimization of the Punzi figure of merit [9]. The MVA curve is obtained by training the algorithm on data without $\overline{H}$ trapped, recording only annihilations from muons, and annihilations recorded during the mixing phase, during which the background is negligible and events are mainly constituted by un-trapped $\overline{H}$

The cosmic background corresponding to the red star is the chosen working point, and the data that will be presented are related to that particular selection criterion. The signal efficiency rate is 75.7(5) %, and the background is 37.4(7) mHz. Although these quoted values are estimated from runs with several thousands recorded events, the actual background during the experimental runs can be different. Due to the presence of electrode heating, a portion of the atoms could annihilate on residual gas. The amount of gas present in the trap is not controlled, hence the background is a free parameter in the analysis, and has not been fixed to the expected value from the MVA selection.

4.2 Signal Model Definition

The signal model has been expressed in terms of a parameter that describes the position of the **cb** and **da** lineshapes, and acts therefore as a *frequency marker* (f_o). The difference of the frequency markers for the **cb** and **da** transitions, corrected for the magnetic field decay, corresponds to an estimate of the antihydrogen hyperfine splitting frequency $a_{1S}/h = f_o^{(cb)} - f_o^{(da)}$.

In the analysis model, it is used for the *frequency marker* a definition related to the onset of the transition. With respect to previous measurement [1], however, some modifications are necessary, to account for the increased $\overline{H}$ population in each run (which may result in detectable background from anti-atoms escaping the trap for reasons different than microwave-induced spin-flips) and for the increased sensitivity to differences between the **cb** and **da** lineshapes.

The lineshape model $f(\cdot)$ for the fit of the experimental data is constructed as the

convolution of two functions: a bare function $g(\cdot)$ and a resolution/broadening function $R(\cdot)$:

$$f \equiv g \otimes R. \quad (4.1)$$

The "bare" function is introduced to describe the energy distribution, magnetic field profile and depletion effect which determine the lineshape. It has a well defined onset of the transition, and an exponential decaying right tail. The resolution function has been introduced to describe all those effect that are related to broadening of the resolution conditions, such as magnetic field inhomogeneities, magnetic field fluctuations, Doppler broadening and transit time broadening. The actual shape of g can't be derived directly from first principles, since it would require knowledge of microwave standing wave pattern, actual power delivered to the trap and energy distribution of the atoms. These quantities are not known with enough precision, hence the signal model corresponds to an empirical function which satisfies the following requirements:

- its value is zero for frequency values lower than f_o , and strictly positive for frequencies higher than f_o ;
- it features a monotonic rise (corresponding to an increase in the resonant volume as the frequency is increased) between f_o and the maximum of the function;
- it features a monotonic decay (associated to the decreasing antihydrogen population as the scan progresses) after the maximum.

The "bare" function is described in the following 4.2:

$$g(\bar{x}) = \begin{cases} 0 & \bar{x} < -\sigma(k+1) \\ \exp\left(-\frac{1}{2}\right) \times \left[\frac{\bar{x} + \sigma(k+1)}{\sigma k}\right]^k & [\bar{x} > -\sigma(k+1)] \wedge (\bar{x} < -\sigma) \\ \exp\left(-\frac{\bar{x}^2}{2\sigma^2}\right) & (\bar{x} > -\sigma) \wedge (\bar{x} < 0) \\ \exp\left(-\frac{\bar{x}^2}{2\sigma_r^2}\right) & (\bar{x} > 0) \wedge (\bar{x} < \sigma_r) \\ \exp\left(-\frac{1}{2}\right) \left[\frac{\bar{x} + \sigma_r(n-1)}{\sigma_r n}\right]^{-n} & \bar{x} > \sigma_r \end{cases} \quad (4.2)$$

The onset (our chosen frequency marker) is defined as the minimum value at which the above equation differs from zero, while the function parameters have the following meaning:

- $\bar{x} \equiv x - x_{max}$. Therefore, $\bar{x} = 0$ is the position of the maximum;
- σ is the standard deviation for the left-side Gaussian-like component;
- k is the degree of the polynomial describing the rise portion;
- $r = \sigma \cdot (k+1)$ is the distance (rise length) between the onset and the maximum;
- σ_r is the standard deviation of the right-side Gaussian core portion, while n describes the power-law right-tail (Crystal Ball function).

To gain some physical intuition on the model, one can think of the 'bare' function g as describing the lineshape we would observe if the atoms were stationary, the microwave radiation monochromatic, the magnetic field perfectly stable, etc., i.e., if all broadening sources were turned off.

In this hypothetical case, for an up-ward microwave frequency ramp, the onset would be well defined: it would correspond to the infinitely-large sample-size limit of the minimum value corresponding to the first significant annihilation count, and would be associated to the lowest value of the magnetic field. In practice, since the experiment is not dealing with stationary atoms and perfect magnetic field, a certain level of broadening is expected.

The resolution/broadening function

Simulations show that spin-flip transitions can occur at frequencies lower than the frequency corresponding to the magnetic field minimum, due to effects such as:

- fluctuations in the magnetic field;
- width of the microwave radiation in the trap;
- Doppler-shift broadening;
- transit-time broadening;
- power broadening.

Estimating the size of these effects from simulations or calculations is expected to be an easier task than simulating the full lineshape, at least under a set of reasonable simplifying assumptions. Therefore, we describe such broadening effects in the model by convolving the bare function with the resolution function R . The parameters of the resolution function are determined from simulations and calculations, and are fixed in the fit to the $\overline{H}$ data, i.e., there are no floating parameters associated to the resolution function in the fit to the data.

Since monotonic upward frequency scans are performed in the experimental runs, the broadening function is one-sided. Its shape and width can be varied to determine systematic uncertainties associated with the assumptions. The simulations were used to test different combination of bare function g and broadening function R .

The broadening function might depend on the step in the scan (i.e., might vary with frequency), but we can assume that the only part affecting the zero frequency marker is the one evaluated at the first significant step.

The broadening may also depend on the population kicked out in the first step, which might in principle differ between the two transitions.

Even if the two broadening functions were the same, a different underlying bare model would result in a different effect on the *frequency marker*, which wouldn't cancel in the difference between the two transitions.

We can assume different shapes for the broadening function, such as a one-sided Gaussian or a one-sided cusp. A right-sided tail (of negligible width, $t_R = 0.2$ kHz, compared to the frequency step) is included for numerical stability. For a Gaussian we have:

$$R(\overline{x}) = \begin{cases} \exp\left(-\frac{1}{2} \frac{x^2}{t_{gauss}^2}\right) & x < 0 \\ 0 & x \geq 0 \end{cases} \quad (4.3)$$

while for a one-sided cusp:

$$R(\bar{x}) = \begin{cases} \frac{1}{t_{cusp}} \exp\left(-\frac{x}{t_{cusp}}\right) & x < 0 \\ 0 & x \geq 0 \end{cases} \quad (4.4)$$

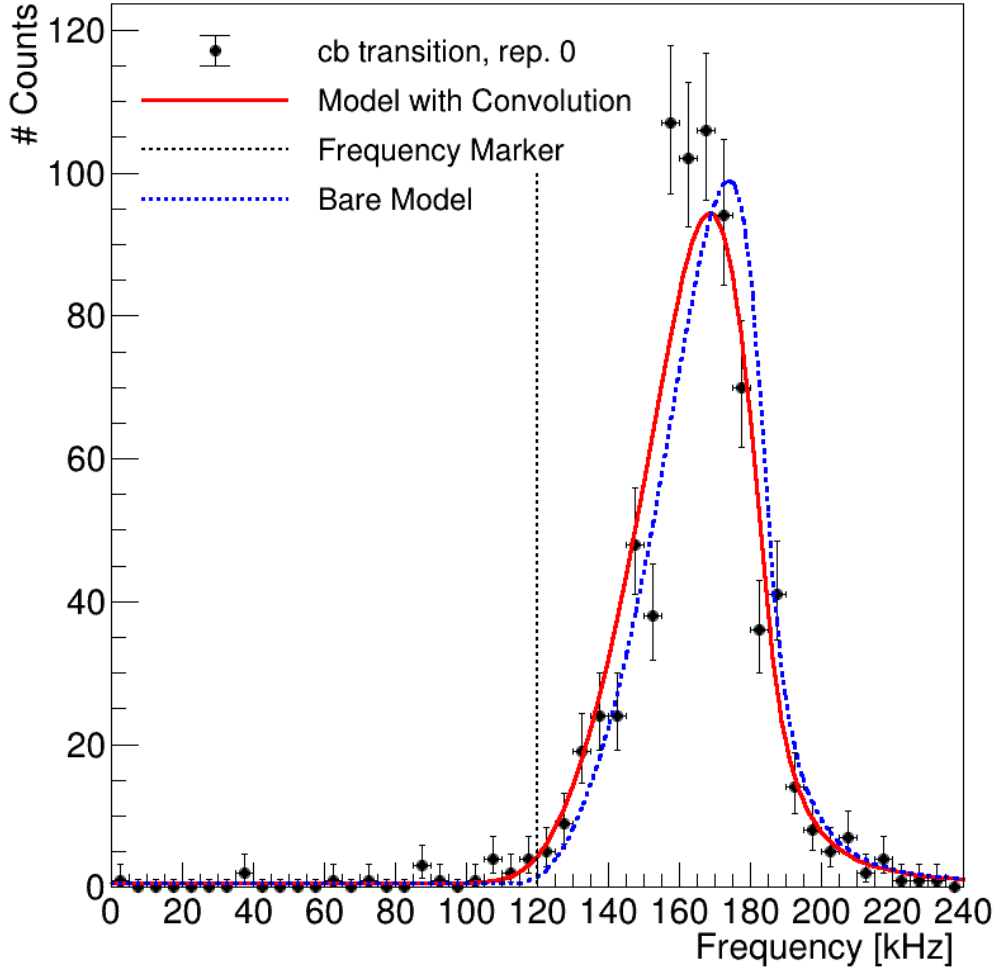


Figure 4.2: Lineshape of the first repetition of the high field dataset. The blue line represents the *bare* model with best fit parameters. The red line represents the *bare* model convoluted with the resolution function (asymmetric gaussian, with 10 kHz), the black vertical dotted line is the frequency marker or onset position.

4.3 Simulation

In this section explain the simulation and how it has been used to estimate the systematic. The shown the plot of the simulation lineshapes and describe the limits of the simulation.

Simulation of the experiment, involving antihydrogen motion and its interaction with microwaves have been conducted to inform the model parameters and study some of the systematic of the measurement. The simulation reflect the experimental protocol following the magnets manipulation and the frequency ramps. The magnetic field has been tuned using ECR data, the ones shown in 3.4

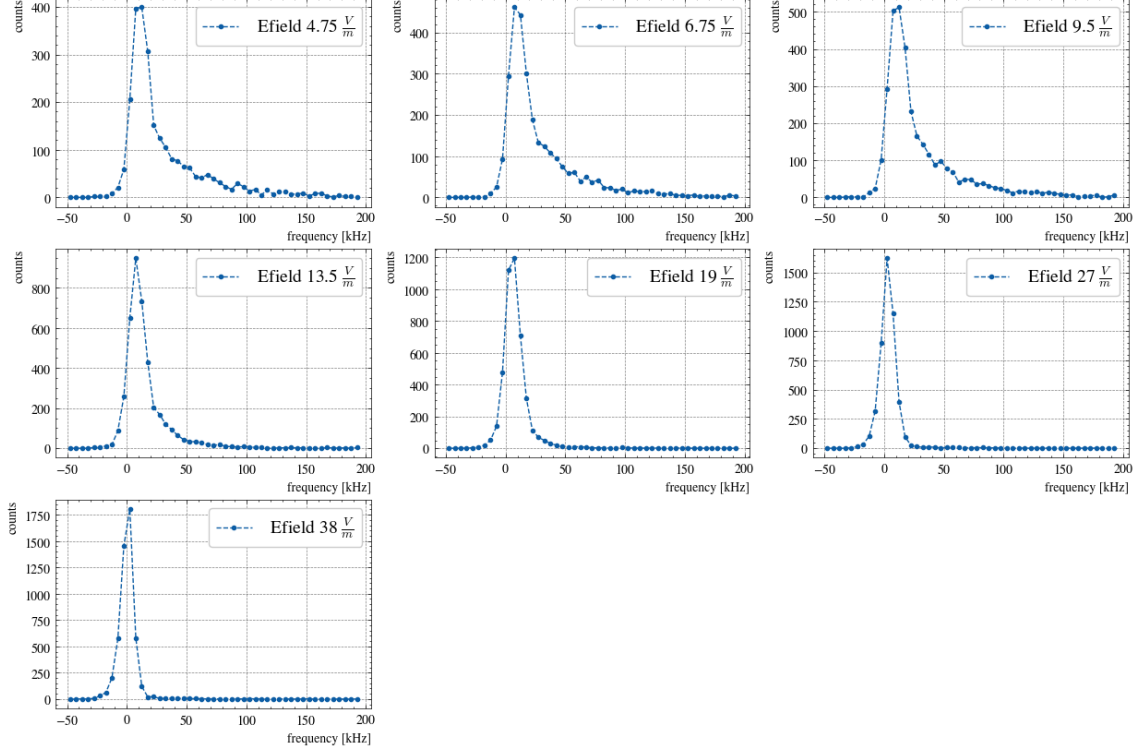


Figure 4.3: Simulation of microwave induced annihilation for different electric field amplitudes (in legend reported as V m^{-1}). The simulation does not account for differences in the standing wave pattern for the **cb** and **da** transitions; consequently the simulated the **da** is essentially identical to the **cb** one, aside from the frequency offset corresponding to the splitting.

The simulation can not describe the experiment with required precision because of the lack of knowledge of the magnetic field off-axis and the unknown microwave field pattern in the trap. Therefore the simulation has been used to inform the signal model parameters, such as degree of the power-law rise of the lineshape. Different lineshapes corresponding to different microwave power are available, and have been fitted with the current signal model.

4.4 Maximum Likelihood Fit

The analysis approach to extract the splitting from the raw data has been developed using the frequentist framework, where the best-fit parameters are resulting from the maximization of the likelihood function $\mathcal{L}$. As mentioned in the previous chapter 3.3, the data are divided in two set of eight repetitions, consisting on a single **cb** and **da** transition measurement interspersed with stacking phases. Details of the specific runs and available data are reported in table 4.1.

Series 7				
Carlsberg reset	19 May 2024: reset to 194.358 A			12:40
Repetition	Run	cb Init Freq (kHz)	Unix Time (s)	Time
0	70377	29 418 364	1716123393.873294	14:03
1	70378	29 418 284	1716129652.988256	
2	70379	29 418 187	1716135002.727111	
3	70380	29 418 073	1716140853.490512	
4	70381	29 417 956	1716146112.077295	
5	70382	29 417 741	1716155966.214942	
6	70383	29 417 607	1716162057.169858	
7	70384	29 417 488	1716167889.394672	
8 (bkg study)	70385			
post	70386	29 416 964	1716193095.703465	10:48 (+1d)
Series 8				
Carlsberg reset	23 May 2024: reset to 186.426 A			13:13
Repetition	Run	cb Init Freq (kHz)	Unix Time (s)	Time
0	70462	28 227 463	1716497309.208071	22:38
1	70463	28 227 387	1716502378.209453	
2	70464	28 227 231	1716509895.094319	
3	70465	28 227 132	1716514501.176821	
5	70467	28 226 914	1716524509.825337	
6	70468	28 226 811	1716529827.672246	
7	70469	28 226 710	1716534513.851191	
8	70470	28 226 613	1716539621.420382	
pre	70461	28 227 609	1716492680.251725	21:28
post	70471	28 226 227	1716558788.451213	15:53 (+1d)

Table 4.1: List of runs used in the analysis for Series 7 and 8. The initial frequency of the **da** scan is set 1 420 390 kHz above the initial frequency of the **cb** scan.

The time interval between a run and the next one is 1 h during which antihydrogen is accumulated in trap by repeating mixing cycles of 4 minutes each. The total statistic accumulated for each run is 1400 – 1600 atoms, that are equally split between c and d states. The Maximum Likelihood fit consist on analysing the data simultaneously, with a likelihood defined as the sum of the individual likelihood for each repetition:

$$\mathcal{L}_{tot} = \sum_{cb;da} \sum_i \mathcal{L}_i(\mu, \vec{x}) \quad (4.5)$$

Where $\vec{\mu}$ is the parameter space, $\vec{x}$ are the observations. The total likelihood is the sum of the likelihood of each individual spectra, and the minimization acts on the total likelihood. The advantage of this shared minimization is that the model complexity can be reduced, since some of the parameters are shared among repetitions of the same transition, increasing the statistical power for fit of the shape parameters, improving the fit minimization. In the nominal analysis, the parameters controlling the shape of the spectra (σ, σ_r, n_r) are shared among repetition of the same transition, while the yields nS_i , the background and the frequency markers $f_{cb,da;i}$ are individual for each repetition.

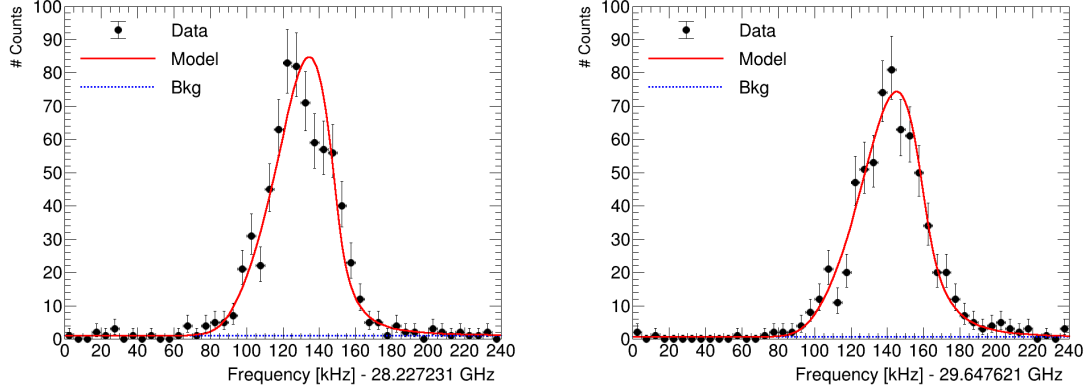


Figure 4.4: Result of the maximum likelihood fit on repetition 3 for the set at 1.03 T.

4.4.1 Blinding the Analysis

The analysis was conducted blind to avoid bias during the development. A random offset value was added to all frequency markers $f_{d \rightarrow a}$ transition, applied uniformly across both datasets. Consequently, the hyperfine splitting extracted from both series was shifted by the same unknown quantity. This offset was revealed only when all the systematic uncertainties had been determined and a common consensus of the analysis procedure had been reached among ALPHA collaboration members.

4.5 Correlation between Observable

The combined *MLE* fit approach, used to analyze the spectra offers several advantages. First, it reduces model complexity by allowing certain parameters to be shared across different datasets rather than independently fitted for each. Second, shape parameters like σ_{rise} and n_r benefit from the increased statistical power, as a consequence of being treated as global parameters. On the other hand, this approach introduces correlations between parameters -in particular the frequency markers- that must be carefully accounted for when deriving statistical uncertainties. Figure 4.5 shows the correlation heat map of the *MLE* fit of the low-field series. As evident from the map, σ_{cb} and σ_{da} which controls the onset shape of the spectra are correlated with the frequency markers. This correlation arises because the shape parameters are constrained to be identical across repetitions, which in turn constrains the fitter frequency markers.

4.6 Systematic Uncertainties

4.6.1 Bare Function

Parameters controlling the Signal Shape

The shape parameters obtained from the fit to simulation fit, which rules the power law rise and the broadening, are used in the nominal *MLE* fit to extract the hyperfine-splitting. However, since the simulation does not have the complete physics to fully describe the

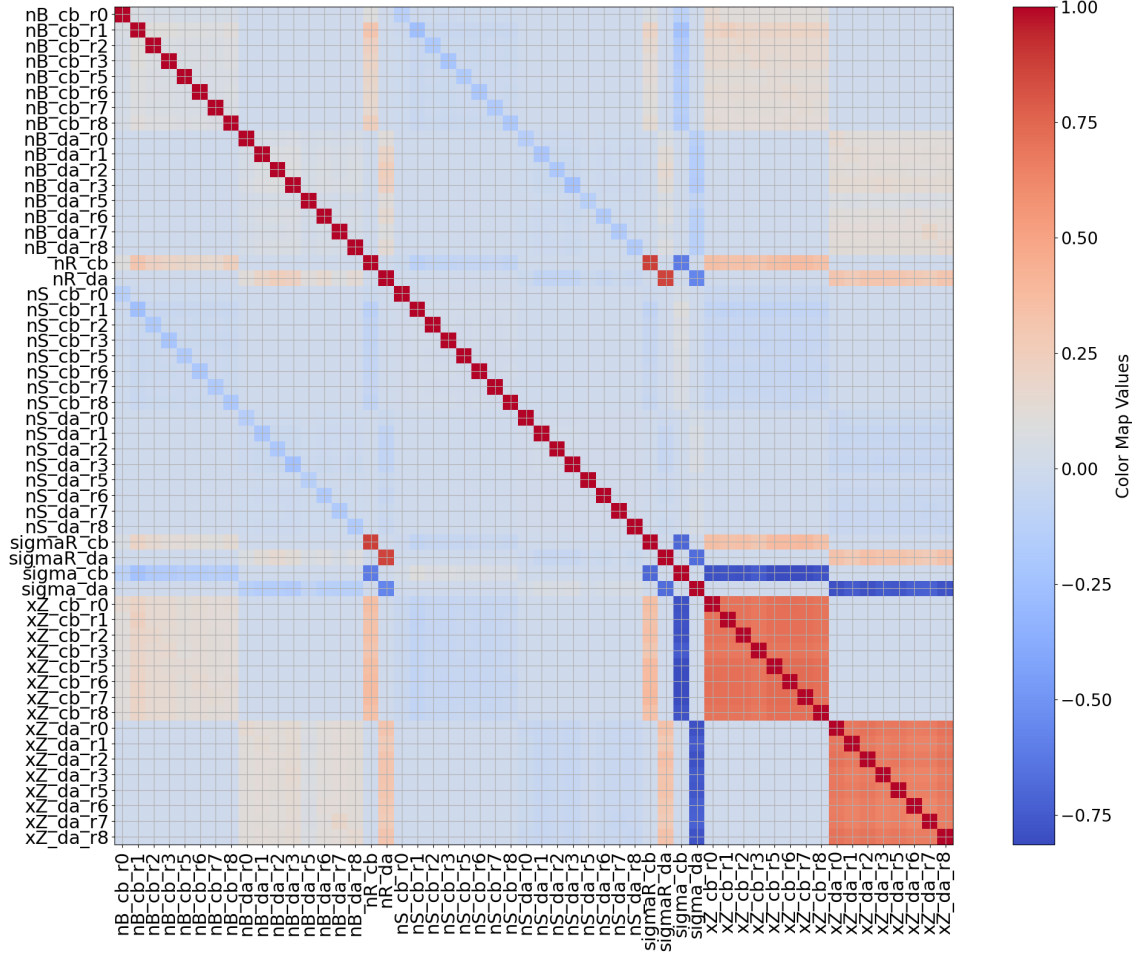


Figure 4.5: Correlation Matrix of the *MLE* fit for the set of data at 1.03 T. The block of parameters identified as $xZ_x x_r y$ (xx denote the transition type, y the transition number) corresponds to the frequency markers (the nomenclature used in the analysis code). It's evident the correlation of the frequency markers within the same transition, as their correlation with the shape parameter σ_{xx} . Correlations are also observed among others shape parameters $\sigma_{R,xx}$ and $n_{r,xx}$. These do not pose a problem since they influence the depletion tail of spectrum, and are insensitive to onset position.

	f_{cb}								f_{da}							
f_{cb}	2.23	1.36	1.34	1.41	1.40	1.40	1.31	1.40	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	1.36	2.16	1.33	1.39	1.38	1.39	1.29	1.39	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	1.34	1.33	2.23	1.38	1.37	1.38	1.29	1.38	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	1.41	1.39	1.38	2.32	1.44	1.44	1.35	1.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	1.40	1.38	1.37	1.44	2.37	1.43	1.33	1.43	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	1.40	1.39	1.38	1.44	1.43	2.13	1.34	1.44	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	1.31	1.29	1.29	1.35	1.33	1.34	2.11	1.34	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
	1.40	1.39	1.38	1.44	1.43	1.44	1.34	2.28	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
f_{da}	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	2.36	1.50	1.46	1.48	1.40	1.44	1.43	1.51
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.50	2.04	1.39	1.41	1.33	1.37	1.36	1.44
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.46	1.39	2.16	1.38	1.31	1.34	1.33	1.40
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.48	1.41	1.38	2.14	1.32	1.36	1.34	1.42
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.40	1.33	1.31	1.32	2.11	1.29	1.28	1.35
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.44	1.37	1.34	1.36	1.29	1.88	1.31	1.39
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.43	1.36	1.33	1.34	1.28	1.31	1.97	1.37
	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	1.51	1.44	1.40	1.42	1.35	1.39	1.37	2.16

Table 4.2: Covariance Matrix, values in off-diagonals blocks are on the order of 10^{-4} / 10^{-5} , and the values have been written with a precision of 2 decimal places.

spectra, a simultaneous fit to the available high statistic sample was performed, to benchmark the result of the simulation and to evaluate systematic uncertainty associated with the fixed parameters used in the nominal procedure. In the current dataset used for this analysis three high statistics repetitions were taken. Two of them for low magnetic field series (one before the 8 repetitions, 1 after the 8 repetitions) and one for the high field series (after the nominal 8 repetitions).

The MLE fit is different from the nominal fit. Specifically, the aim is to get an estimation from the data of the shape parameters, in particular k , in order to use that value as an alternative way in the nominal fit to extract the hyperfine splitting. The absolute difference between the nominal extracted value and the alternative analysis with k is then used to quote the systematic uncertainty. Because of this, k is a free parameter, shared among all the high statistic samples. Instead of minimizing the frequency markers individually (using f_{cb} and f_{da} as free parameters), the fit was varied with the intention of fitting f_{cb} and the splitting itself Δ_f , such that the $f_{da} = f_{cb} + \Delta_f$. By doing so, it is possible to impose a common splitting between the three high statistic repetitions too. The result of the fit are shown in table 4.3, while the plot in figure 4.6 shows the fit model and data.

The k result of the fit is 5.4, which has been used to evaluate a component of the systematic uncertainty associated with function shape variation. The high statistic repetitions have been used also for estimating different systematic related to the analysis. Since those data have been used for this task, they are not included in the main analysis, which relies only on the eight repetitions with less statistical power.

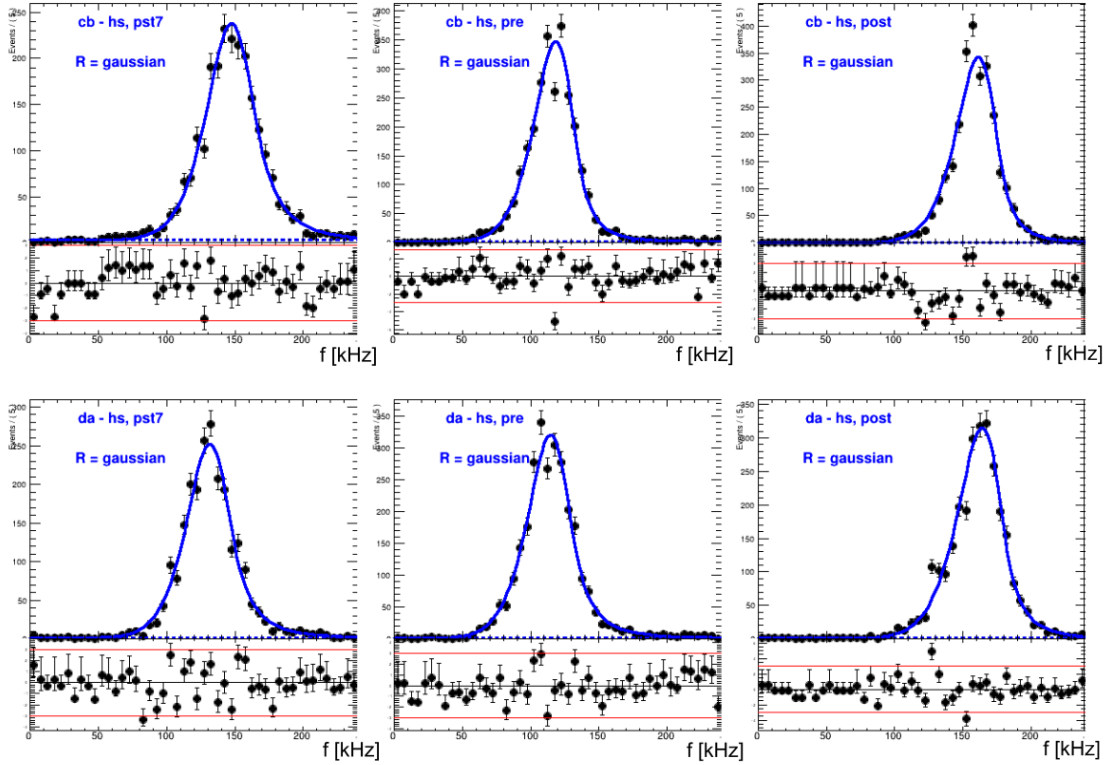
4.6.2 Resolution Function

4.6.3 Reproducibility: Shape variation during the experiment

A possible systematic uncertainty is linked with assumption that the shape of the spectra does not change from repetition to repetition. This assumption is used implicitly in the fit protocol, since the parameter controlling the shape of signal model (σ , σ_r and n_r) are constrained to be the same for all the **cb** and **da** respectively. Hence, a dedicated study is necessary to evaluate this uncertainty.

	Series 7 post	Series 8 pre	Series 8 post
k (common)		5.4 ± 0.2	
$n^{(cb)}$ (common)		7.3 ± 2.4	
$n^{(da)}$ (common)		6.3 ± 1.6	
$\sigma^{(cb)}$ (kHz)	17.0 ± 0.8	15.1 ± 0.5	15.9 ± 0.5
$\sigma^{(da)}$ (kHz)	15.8 ± 0.7	15.7 ± 0.5	17.6 ± 0.5
$\sigma_r^{(cb)}$ (kHz)	14.3 ± 1.0	8.9 ± 0.6	8.7 ± 0.6
$\sigma_r^{(da)}$ (kHz)	12.2 ± 0.8	11.2 ± 0.7	9.9 ± 0.7
$n_B^{(cb)}$	150 ± 22	109 ± 20	33 ± 11
$n_B^{(da)}$	80 ± 16	83 ± 21	62 ± 14
$n_S^{(cb)}$	2271 ± 48	2687 ± 53	2705 ± 52
$n_S^{(da)}$	2221 ± 48	2724 ± 54	2724 ± 52
Δf (kHz)		-12.7 ± 2.8	
time from center of associated series (h)	11.5	-7	11.2
duration of associated series (h)	12	11.75	11.75

Table 4.3: Model float parameters from high statistics sample

Figure 4.6: Fit results on the high statistics samples, with the shared k parameter left free to vary in the fit, and with shared Δf , to help the convergence of the fit.

The variation of the shape of the spectra is something expected during the measurement. There are a number of physical arguments that indicates this. At first, the energy

distribution of the atom ensemble used during the can vary from repetition to repetition (although the anti-atoms are accumulated along 20 mixing cycles, so the average energy fluctuations should be small). More importantly, since the current of the main solenoid is drifting and decreasing over time, the mw frequency must be adjusted to follow the magnetic field drift. Hence, the working point of the frequency scan changes from repetition to repetition. Following that, microwave standing wave pattern can change, and also the effective power delivered to the trap can change. Additionally, although magnetic field profile could change during the measurement, both its on-axis profile (which can be monitored by ECR measurement) and its off-axis profile (which is unmonitored). The magnetic field profile is monitored using a direct-current-current transformers based on magnetometers sensors (ITZ 2000-SB FLEX ULTRASTAB from LEM International SA), and its value is used as input of PIDs that ensures stability of each individual magnet (more details in 4.7.3).

Shape variation are expected to have a non negligible impact on the measurement. As mentioned before, correlation between the shape parameters and the frequency markers is strong (look at heatmap 4.5). Hence, shape variations from repetition to repetition are expected to change consistently the extracted value. For this reason, the experimental protocol incorporated high-statistic reference measurements, that were used to estimate the magnitude of these variations. The data taking comprises an high statistic measurement taken before the eight nominal repetition (only for the low field series), and high statistic measurement after the eight nominal repetition (for both series). Those high statistic sample measurement has a statistical power 5-6 times the one of a single repetition. These data have been fitted independently, with the same model used in the nominal Maximum Likelihood Fit. The parameters resulting from the fit are reported in table 4.4

	Series 7 post	Series 8 pre	Series 8 post
$n^{(cb)}$	4.3 ± 1.7	5.1 ± 0.9	6.4 ± 2.9
$n^{(da)}$	9.8 ± 5.6	3.5 ± 0.9	3.6 ± 1.21
$\sigma_r^{(cb)}$ (kHz)	13.48 ± 1.4	7.3 ± 0.8	8.3 ± 0.8
$\sigma_r^{(da)}$ (kHz)	13.00 ± 1.2	9.0 ± 1.0	7.7 ± 1.1
$\sigma^{(cb)}$ (kHz)	22.0 ± 1.1	20.3 ± 1.0	17.2 ± 0.7
$\sigma^{(da)}$ (kHz)	17.1 ± 0.9	20.03 ± 0.8	24.8 ± 0.8
$n_B^{(cb)}$	186 ± 25	130 ± 26	81 ± 15
$n_B^{(da)}$	125 ± 20	86 ± 22	70 ± 14
$n_S^{(cb)}$	2235 ± 51	2666 ± 56	2658 ± 53
$n_S^{(da)}$	2175 ± 49	2722 ± 55	2716 ± 53
Δf (kHz)	-9.17 ± 2.8	-8.7 ± 2.2	-20.0 ± 2.1
$rise^{cb}$ (kHz)	66 ± 3	60.8 ± 2	52 ± 2
$rise^{da}$ (kHz)	51 ± 3	60.1 ± 3	74 ± 2
time from center of associated series (h)	11.5	-7	11.2
duration of associated series (h)	12	11.75	11.75

Table 4.4: Model float parameters from high statistics sample. The Δf value (not corrected for the B -field drift, and blind) is shown as a cross-check.

These results can be used to evaluate the effect of shape variations on the inferred value of the hyperfine splitting. To estimate such uncertainty, only the *post* distributions

have been used. Assuming a difference in shape in time, the 8 repetitions have been fitted once using the shape model obtained from the *post* distribution and once with the model obtained from the high statistic template itself. Since the *post* were collected about $\Delta t_{post} = 12$ hours after the middle of the Physics *Series*, that lasted about $T_{series} = 12$ hours (see Fig. ??), the difference in the result has been divided by 2 (since the Series extends $\pm T_{series}/2 \approx \pm \Delta t_{post}/2$ around the center of the Series). This estimate resulted in ~ 4 kHz for the Series 7 and ~ 7 kHz for the Series 8. The best-fit parameters for the high statistics sample are reported in Tab. 4.4.

Another cross check has been performed comparing the result of the HFS measurement, once imposing a common rise (parameter r in Eq. 4.2) for the 8 repetitions (the way it is done for the measurement) and once leaving such parameter to vary across the 8 repetitions (see Fig. ??) as described in Sec. ??.

4.6.4 Data Selection and Binning

Title is self-explaining

4.7 Magnetic Field Drift

In 3.3.2 we have introduced an important effect that requires treatment in the analysis: the decay of the bias (Carlsberg) magnetic field during the course of the experiment. Due to the delay of the measurement for the higher-frequency transition in each repetition, a correction must be applied to the hyperfine value of the hyperfine splitting: this correction can be directly inferred from data. In the analysis it is assumed that the magnetic field decay acts linearly, and with the same slope value, for the two transition. Hence, the two sets of 8 frequencies plotted versus time will lie on two parallel lines. As a consequence, it is possible to derive a correction associated to the field decay from a linear fit to the frequency markers. The linearity assumption has been relaxed, a different approach has been followed to estimate deviations from linearity, using GPR¹ interpolation, only assuming that the hyperfine splitting for antihydrogen does not depend on the bias field in the range of this measurement campaign.

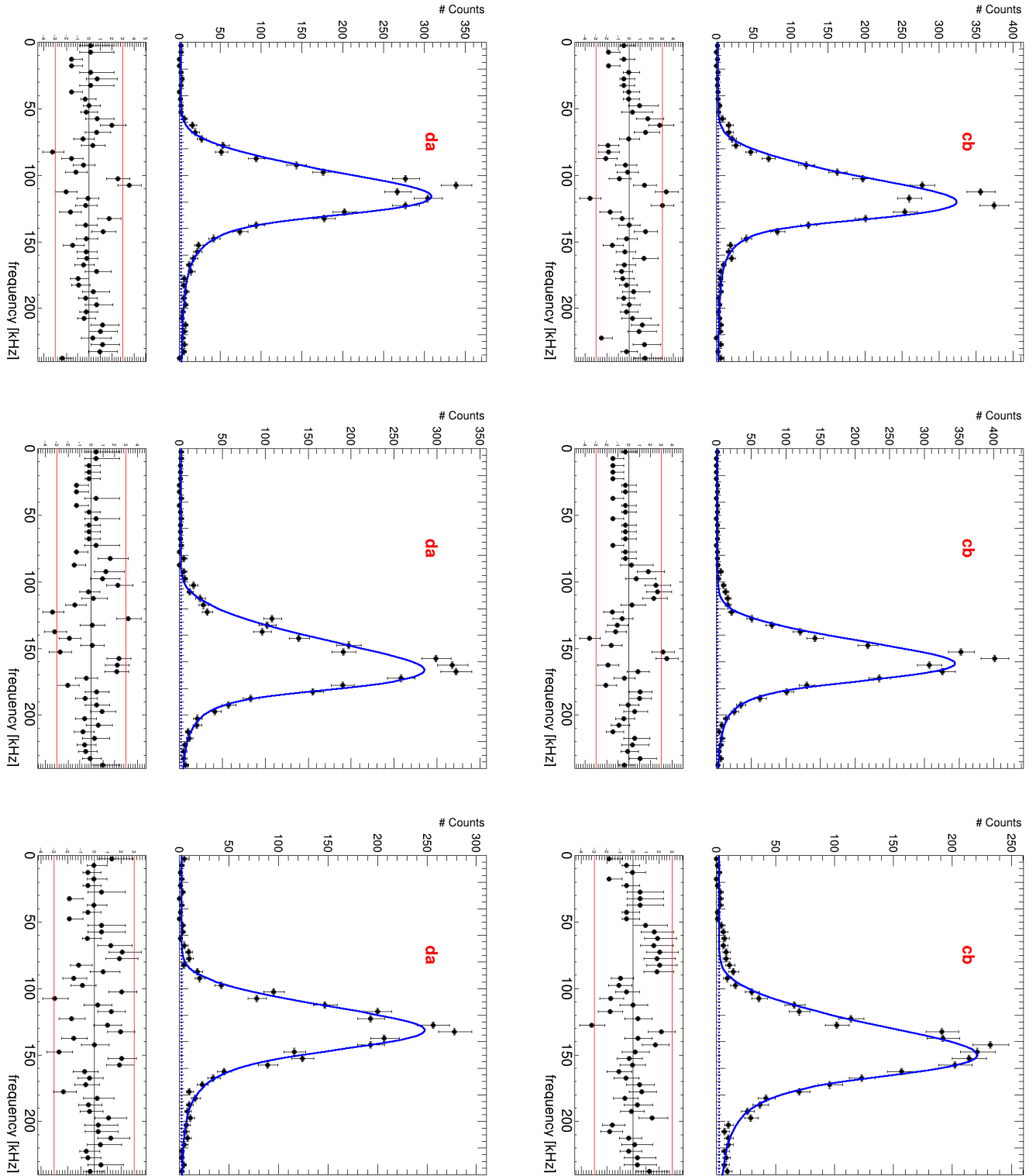
4.7.1 Extended Least Square Minimization

The nominal procedure assumes the decay of the solenoidal magnetic field to be linear in time and equal for the two transitions. The hyperfine splitting $\nu_{HF}^{unblind}$ is extracted with a linear fit to the 8 **cb** and 8 **da** frequency markers (after having set $f_{blinding}$ to 0), imposing a common slope for both transitions.

As can be noted from the correlation matrix reported in Tab. 4.2, the frequency markers, that correspond to the y_i observables, are correlated between each other, due to the presence of shared parameters in the combined fit to the repetition datasets. The algorithm used for the linear fit is the Generalized Least Square (GLS), a generalization of the ordinary least square algorithm. This generalization is necessary to properly account for correlations. The linear model can be expressed as:

$$\vec{y} = \vec{\beta} \mathbf{X} + \vec{\epsilon} \quad (4.6)$$

¹GPR: gaussian process repressor



where the $\mathbf{X}$ matrix contains the time of each repetition; β is the fit parameter vector, ϵ is the error vector of the y_i observables, which is described by the covariance matrix Ω reported in Tab. 4.2.

The above equation can be explicitly written in matrix form, to highlight the arrangement of the variables for the combined fit procedure:

$$\begin{bmatrix} f_{cb}^0 \\ f_{cb}^1 \\ \vdots \\ f_{da}^0 \\ f_{da}^1 \\ \vdots \end{bmatrix} = \begin{bmatrix} 1 & -0.5 & t_{cb}^0 \\ 1 & -0.5 & t_{cb}^1 \\ \vdots & \vdots & \vdots \\ 1 & +0.5 & t_{da}^0 \\ 1 & +0.5 & t_{da}^1 \\ \vdots & \vdots & \vdots \end{bmatrix} \times \begin{bmatrix} q \\ hfs \\ m \end{bmatrix} \quad (4.7)$$

The solution for the regression problem is given by the following equation, whose demonstration is omitted (for reference: <https://www.biostat.jhsph.edu/~iruczins/teaching/jf/ch5.pdf>):

$$\vec{\beta} = (\mathbf{X}^T \Omega^{-1} \mathbf{X})^{-1} \mathbf{X}^T \Omega^{-1} \vec{y} \quad (4.8)$$

$$\sigma_{\beta}^2 = (\mathbf{X}^T \Omega^{-1} \mathbf{X})^{-1} \quad (4.9)$$

4.7.2 Short Term Upward Drift

The long term linear drift is not the only effect requiring careful treatment. A short-term upward drift may also occur during the experiment. This is expected because the accumulation and stacking phase between repetitions involves changes to the beam-line magnets. Although the main solenoid magnets and Mirror coils currents are not changed during this operation, the beam-line magnets must be changed, for the plasma manipulation and transport from the catching region to the atom trap. Parasitic currents could be induced in the solenoid through mutual coupling, producing a magnetic field shift in the magnetic field distinct from the linear. Since the experiment is based on a **cb** transition followed by **da** transition, an induced shift in the magnetic field is directly related to a systematic shift of all the **da** transitions due to the Zeeman effect. Hence, a systematic shift in the splitting of the transition is expected, that is unrelated to the physical splitting.

In principle, this effect can be estimated through theoretical calculation of the parasitic currents induced from the beam-line magnets to the Mirror Coils and the superconducting solenoid. In practice, this approach is challenging, given the required target accuracy of this measurement. So a data driven approach was followed.

The systematic uncertainty associated with this effect has been constrained using data taken on 2023, following an experimental protocol similar to the one used for the 2024 campaign, but with specific modification to isolate and study this effect. The protocol consisted of repeating the same measurement, a **cb** and **da** transition pair, with a deliberate delay introduced in the sequence between the final manipulation of the beam-line currents and the start of spectroscopy. In the first variant, the spectroscopy began 32s after the last manipulation of the beam-line magnet change. In the second variant, the delay was set to 320s. The delays were chosen in such a way that the **cb** measurement

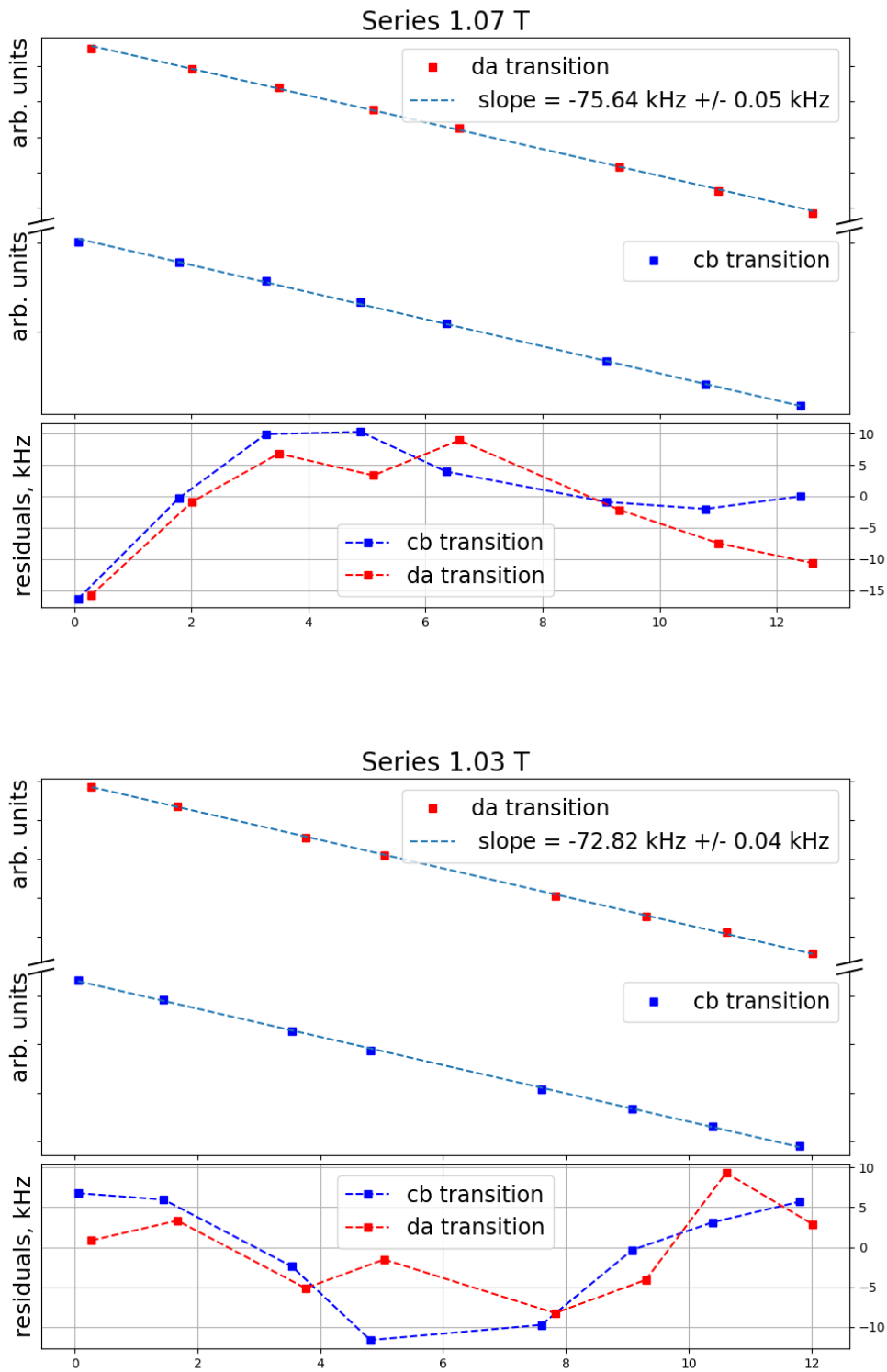


Figure 4.7: •

with the second protocol coincides in time with the **da** measurement of protocol 1. and the data taking consisted of alternately repeating one measurement with protocol 1 and one with protocol 2. Apart from this modification, the data taking procedure were analogous to those employed in 2024. 6 repetitions with the first protocol and 5 repetition with the second were taken, to correct the linear magnetic field drift and extract the transition splitting. In following analysis, we denote $cb_{32s(352s)}$; $da_{32s(352s)}$ as the frequency marker intercepts from the linear fit corresponding to the first (second) protocol. The frequency shifts induced by the short-term upward magnetic drift can then be expressed as follows:

$$\begin{aligned}\Delta_{cb} &= cb_{32s} - cb_{352s} \\ \Delta_{da} &= da_{352s} - da_{32s}\end{aligned}$$

Three combinations of splitting can be determined from the analysis of the experimental data. The spectra were fitted using the same lineshape function as in 2024 analysis, and the fit to the frequency markers has been performed with a slope constrained to be in common to all the four lines (see Figure 4.8). Because the **cb** for protocol 2 and **da** for protocol occur at the same time relative to last magnet manipulation, it follows thatL:

$$hfs = da_{32s} - cb_{352s}$$

The other two quantities of interest are:

$$\begin{aligned}da_{352s} - cb_{352s} &= da_{32s} + \Delta_{da} - cb_{352s} = hfs + \Delta_{da} \\ da_{32s} - cb_{32s} &= da_{32s} - \Delta_{cb} - cb_{352s} = hfs - \Delta_{cb}\end{aligned}$$

The linear fit was performed with the same minimization technique (extended least-square), with a common slope m , the hyperfine splitting hfs , common offset q and shifts Δ_{cb} and Δ_{da} as free parameters. The best fit values have been reported in figure 4.8. The 2024 data taking had the **cb** measurement starting 32 s after the beam-line manipulation, and the **da** transition measured after 512 s + 300 s of wait. So the overall shift can be approximated as:

$$da_{2024} - cb_{2024} \simeq da_{352s} - cb_{32s} = hfs - \Delta_{cb} + \Delta_{da} \quad (4.10)$$

From this analysis, the systematic shift due to the short-term upward magnetic shift was estimated as $\sigma_{shift} = |\Delta_{da} - \Delta_{cb}| = 1kHz$. This value has been added in quadrature as a systematic uncertainty to the overall error budget of the measurement.

4.7.3 DCCT Traces and Magnetic Field Fluctuation

The magnets employed to design the microwave flat magnetic field for the measurement are supplied with hundred of amperes, and are handled and monitored with direct current-current transformers based on closed-loop fluxgate magnetometers sensors (DCCT). The current flowing in each magnets is constantly measured by analog-to-digital converter (NI-9239 cRIO) that digitize the DCCT signal output, and it's used in PID control of the magnet power supply for stabilization within few mA. In the context of the analysis, the DCCTs current traces have been analyzed to verify how stable the currents are versus time.

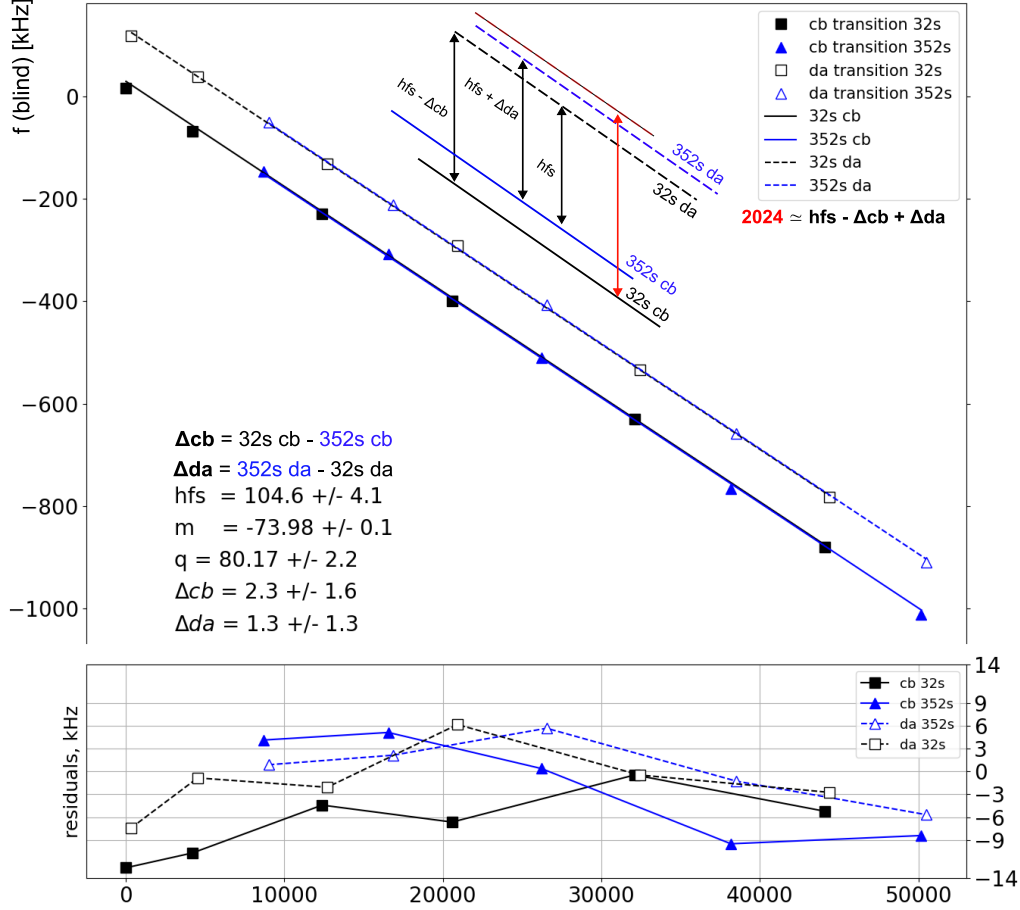


Figure 4.8: This plots shows the results from the analysis of the 2023 data. This analysis has been used to constrain magnetic upward drift induced from beam-line magnet manipulation during antihydrogen stacking phase. The change of the beam-line magnets during stacking can induce parasitic currents on Mirror and Solenoid magnets, causing a small magnetic upward drift. Since the **da** transitions always follows the **cb** transition, this effect can generate a systematic shift in the frequency that change the value of the splitting. This shift has been constrained using 2023 data that were collected with two protocols. In the first protocol there are 32s between the start of the microwave measurement. In the second protocol the time interval is increased to 352s. With this setup, the **cb** transition with the 352s delay starts at the same time, with respect to the beam-line magnet manipulation, of the **da** transition with the 32s delay. The frequency markers shown are the result of the analysis on Pass Cuts data.

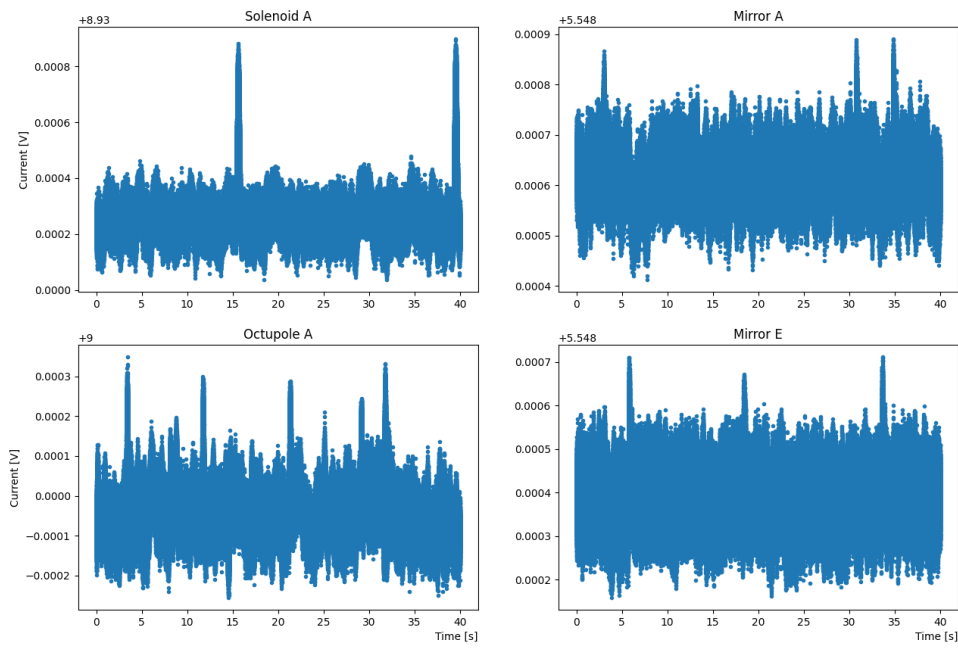


Figure 4.9: Example of DCCTs traces for a total of 40 second of acquisition. The units on the y-axis are Volts, since the DCCT provides a ± 10 V output. The traces have been converted in Ampere units and the amplitude of the glitches has been used to study possible discrepancies on the measurement induced by change in the magnetic field minimum

As evident in 4.9, the DCCTs traces for a selection of magnets show that there are glitches in the magnets currents occurring on a 10 kHz scale. These glitches could potentially influence the magnetic field profile along the axis, which can result in a temporary (fast) change on the resonance frequency of the atoms. Although the duration of the observed glitches for all the magnets has been confirmed short (less than 0.1 s), their amplitudes, when translated in kHz units, was non negligible (order of 10 kHz).

Such an effect can potentially displace and anticipate some of the annihilations that would occur later during the mw scan. Hence, an ad-hoc simulation study was performed to estimate how much such displacement occurring with a certain rate would affect the hyperfine splitting measurement.

The simulation was performed by creating pseudo-data from the real datasets. Glitches similar to the ones observed in 4.9 were simulated with time rate of 10 Hz. Each time a glitch is happening, it is chosen randomly from a list containing all the glitches observed during the 40 s of DCCT traces measured. Each glitch was approximated as a step function, with a duration fixed to be equal to the interval where the glitch goes from 10 percent (before peak) to 10 percent (after peak) and amplitude equal to the observed peak. By considering the peak, we overestimate the displacement of the annihilations.

The simulation consider the displacement to be asymmetrical. This mean that negative current glitches (where the magnetic field decreases) anticipate annihilations happening later-on in the spectra, while positive glitches delayed some of the annihilation to the next bin. With this simulation mechanism, pseudo-data where generated and fitted with the same nominal protocol used in the analysis. The simulation consisted of 40 pseudo-experiments, and the standard deviation of the results was used to estimate the effect of the splitting. The standard deviation of the results obtained for both the series is 0.7 kHz, which is small compared to other systematic uncertainties. Beside this, further consideration indicates that assuming a 100 % conversion from current to frequency displacement is generous overestimation. In fact, because of the coupling between different magnets, the mutual induction contributes to mitigate the effect on the magnetic field profile. This computed value is to be considered as a cross-checks that shows that is such a mechanism is presents, its influence on the measurement is small, hence it was not considered in the final estimation of the systematic uncertainty.

4.7.4 Error Budget Summary

4.8 Validation and Cross-Checks of the Analysis

Chapter 5

Results Summary

5.1 Table of Results

5.2 CPT, outlook

Configuration	Series 7 (kHz)		Series 8 (kHz)		Combination	
	$\nu_{HF}^{blind} \pm \sigma$	Δ	$\nu_{HF}^{blind} \pm \sigma$	Δ	$\nu_{HF}^{blind} \pm \sigma$	Δ
Nominal (“common fit” linear GLS) $k = 2, t_{gauss}^{cb} = 10, t_{gauss}^{da} = 10$	10.83 $\pm$ 1.7	–	8.98 $\pm$ 1.5	–	9.79 $\pm$ 1.1	–
Lineshape reproducibility Nominal with g from post Δ is divided by 2 (see text)	16.48 $\pm$ 0.4	(5.65) 2.8	–4.98 $\pm$ 0.4	(–13.96) –7.0	4.41 $\pm$ 0.3	–5.38 –2.7
Bare function (choice)						
Alt. bare model, $k = 2, t_{cusp} = 6$	11.28 $\pm$ 1.4	0.5	9.65 $\pm$ 0.4	0.7	10.36 $\pm$ 0.4	0.6
(parameters)						
$k = 1$	8.32 $\pm$ 1.0	–2.5	9.31 $\pm$ 0.8	0.3	8.88 $\pm$ 0.6	–0.9
$k = 5.4$	10.70 $\pm$ 3.8	–0.1	10.01 $\pm$ 3.2	1.0	10.31 $\pm$ 2.4	0.5
max uncertainty increase	($\sqrt{3.8^2 - 1.7^2}$)	3.4	($\sqrt{3.2^2 - 1.5^2}$)	3.0	($\sqrt{2.4^2 - 1.1^2}$)	2.1
Resolution function (choice)						
$t_{cusp}^{cb} = t_{cusp}^{da} = 5.1$	11.15 $\pm$ 1.7	0.3	8.99 $\pm$ 1.4	0.0	9.94 $\pm$ 1.1	0.2
$t_{gauss}^{cb} = t_{gauss}^{da} = 13.6$	11.00 $\pm$ 1.9	0.2	9.22 $\pm$ 1.6	0.2	10.00 $\pm$ 1.2	0.2
$t_{gauss}^{cb} = t_{gauss}^{da} = 7.0$	10.9 $\pm$ 1.6	0.1	9.62 $\pm$ 1.4	0.6	9.83 $\pm$ 1.1	1.1
(frequency marker)						
$t_{gauss}^{cb} = 10, t_{gauss}^{da} = 8$	7.64 $\pm$ 1.7	–3.2	5.87 $\pm$ 1.5	–3.1	6.64 $\pm$ 1.1	–3.2
$t_{gauss}^{cb} = 10, t_{gauss}^{da} = 12$	14.34 $\pm$ 1.8	3.5	12.35 $\pm$ 1.5	3.4	13.22 $\pm$ 1.2	3.4
$t_{gauss}^{cb} = 8, t_{gauss}^{da} = 10$	13.75 $\pm$ 1.7	2.9	12.45 $\pm$ 1.4	3.5	13.02 $\pm$ 1.1	3.2
$t_{gauss}^{cb} = 12, t_{gauss}^{da} = 10$	7.43 $\pm$ 1.8	–3.4	5.74 $\pm$ 1.5	–3.2	6.48 $\pm$ 1.2	–3.3
$t_{gauss}^{cb} = t_{gauss}^{da} = 0$	11.97 $\pm$ 1.6	1.1	9.89 $\pm$ 1.3	0.9	10.80 $\pm$ 1.0	1.0
B-drift (fit procedure)						
“split” linear	8.49 $\pm$ 1.7	–2.3	9.51 $\pm$ 1.5	+0.5	9.06 $\pm$ 1.1	–0.7
(long time scale linearity)						
“split” linear vs. “split” GPR	8.24 $\pm$ 1.7	–0.3	9.43 $\pm$ 1.5	–0.1	8.91 $\pm$ 1.1	–0.2
(short time scale drift)						
2023 specific data	N/A	1.1	N/A	1.1	N/A	1.1
Data selection and binning						
Annihilation time shifted by 30 ms	10.84 $\pm$ 1.7	0.0	8.98 $\pm$ 1.5	0.0	9.79 $\pm$ 1.1	–
Binning	N/A	2.0	N/A	2.0	N/A	2.0
Cross checks						
Nominal without $\mu\omega$ transition veto	10.93 $\pm$ 1.7	0.1	8.89 $\pm$ 1.5	–0.1	9.78 $\pm$ 1.1	–

Table 5.1: Results of the nominal maximum likelihood fit. **The results are blind** (see equation ??). For the alternative configurations used to evaluate the systematic uncertainties, the difference Δ with respect to the results of the **nominal fits** are reported. The Δ values shown in **bold**, are the ones summed in quadrature to obtain the total systematic uncertainty reported in Tab. ??.

Appendices

.1 Calculation of the Fine Structure for the 1s Energy Level

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